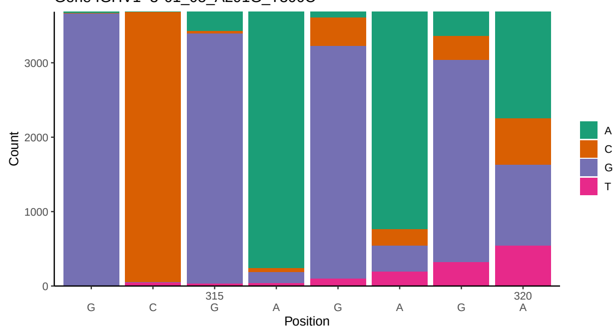
OGRDBstats Report

Contents

1 Novel sequence analysis

1.1 End-nucleotide composition

Gene IGHV1−3*01_05_A291G_T300C



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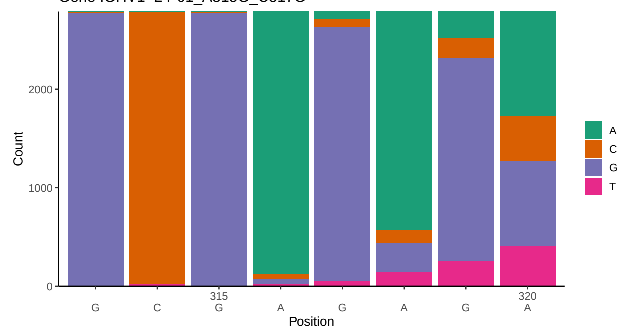

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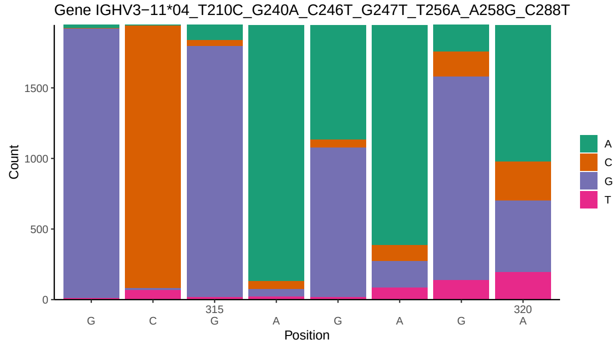

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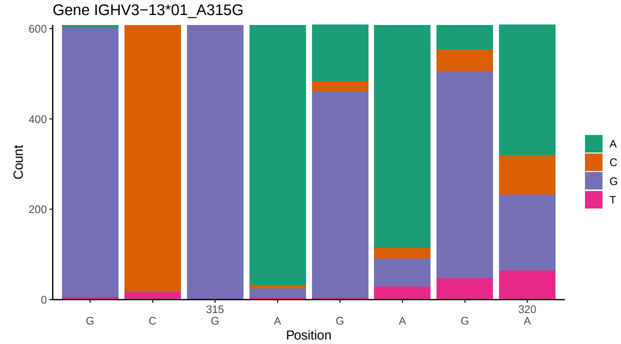

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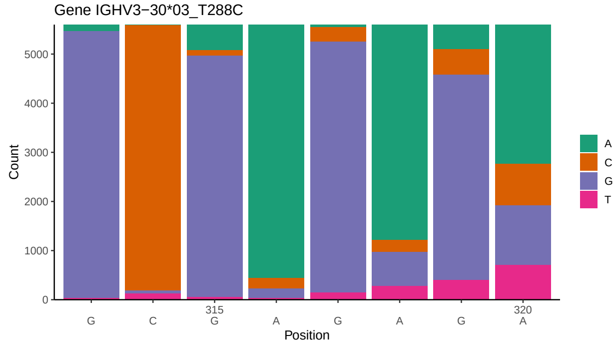

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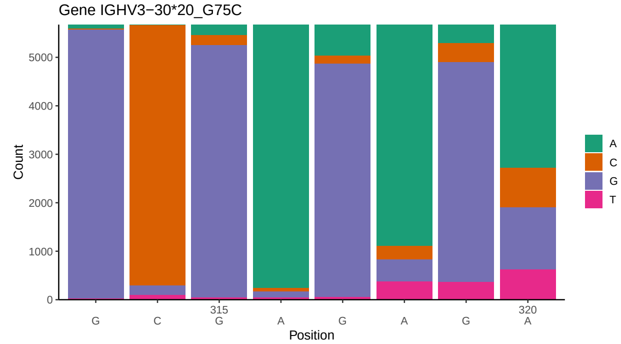

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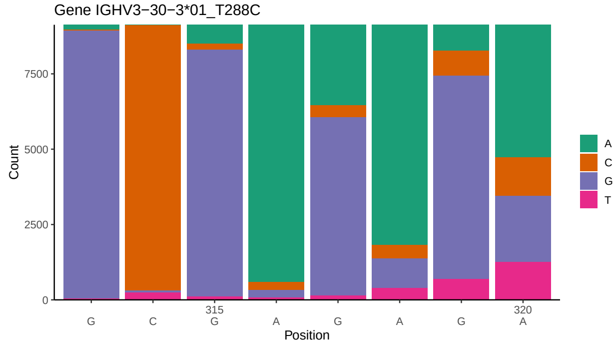

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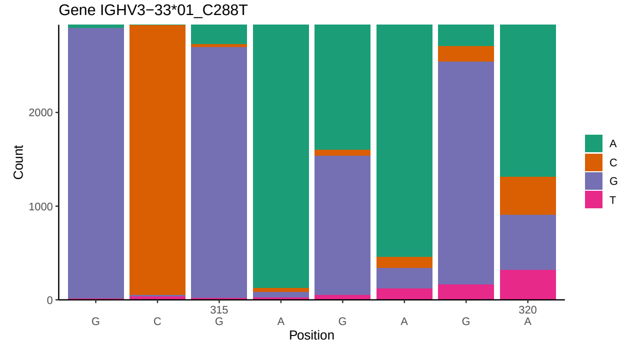

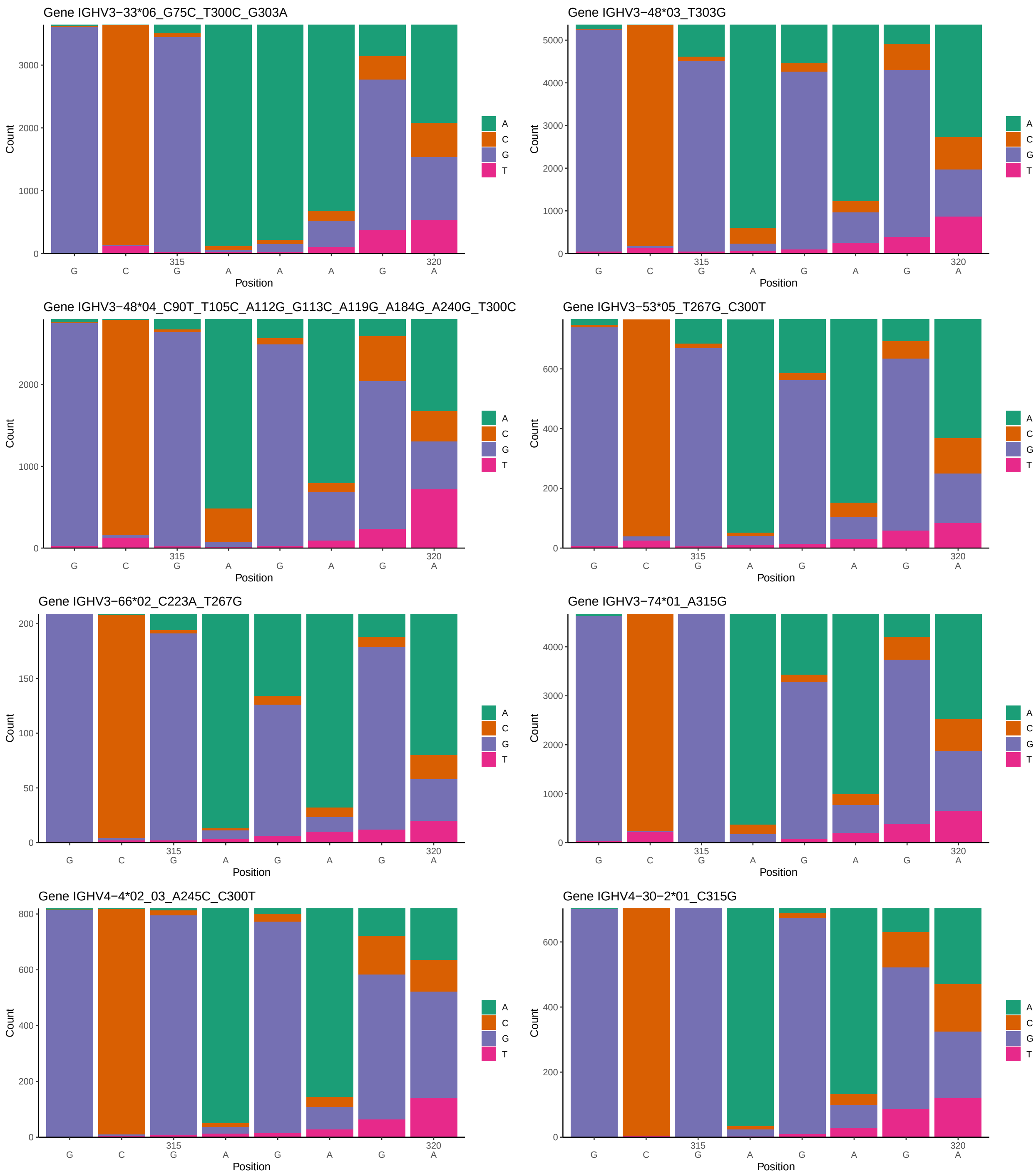
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Gene IGHV3−48*03_T303G
Gene IGHV3−48*04_C90T_T105C_A112G_G113C_A119G_A184G_A240G_T300C
Gene IGHV3−53*05_T267G_C300T
Gene IGHV3−66*02_C223A_T267G
Gene IGHV3−74*01_A315G
Gene IGHV4−4*02_03_A245C_C300T
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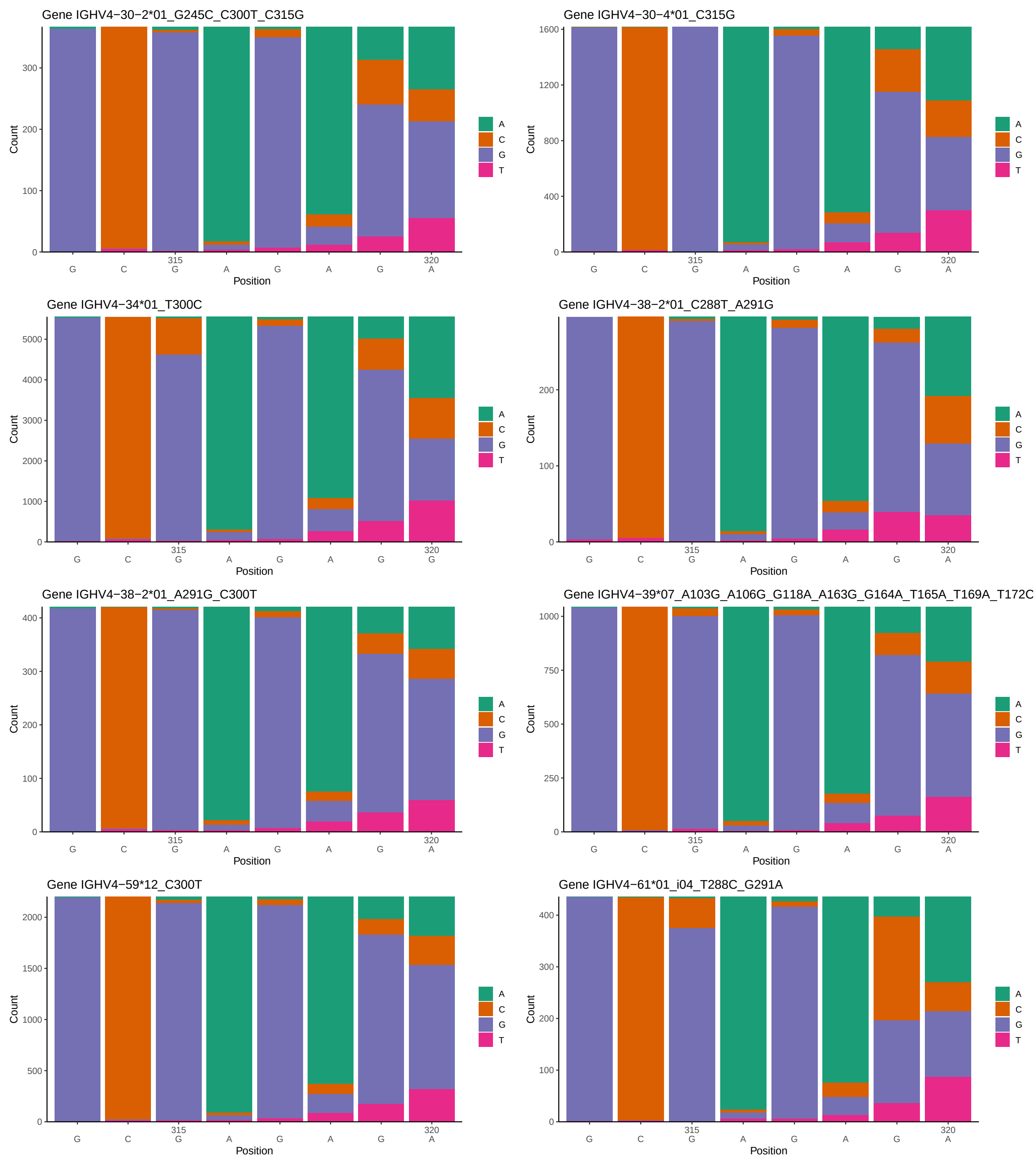
A
C
G
T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Gene IGHV4−30−4*01_C315G
Gene IGHV4−34*01_T300C
Gene IGHV4−38−2*01_C288T_A291G
Gene IGHV4−38−2*01_A291G_C300T
Gene IGHV4−39*07_A103G_A106G_G118A_A163G_G164A_T165A_T169A_T172C
Gene IGHV4−59*12_C300T
Gene IGHV4−61*01_i04_T288C_G291A
Count
Position
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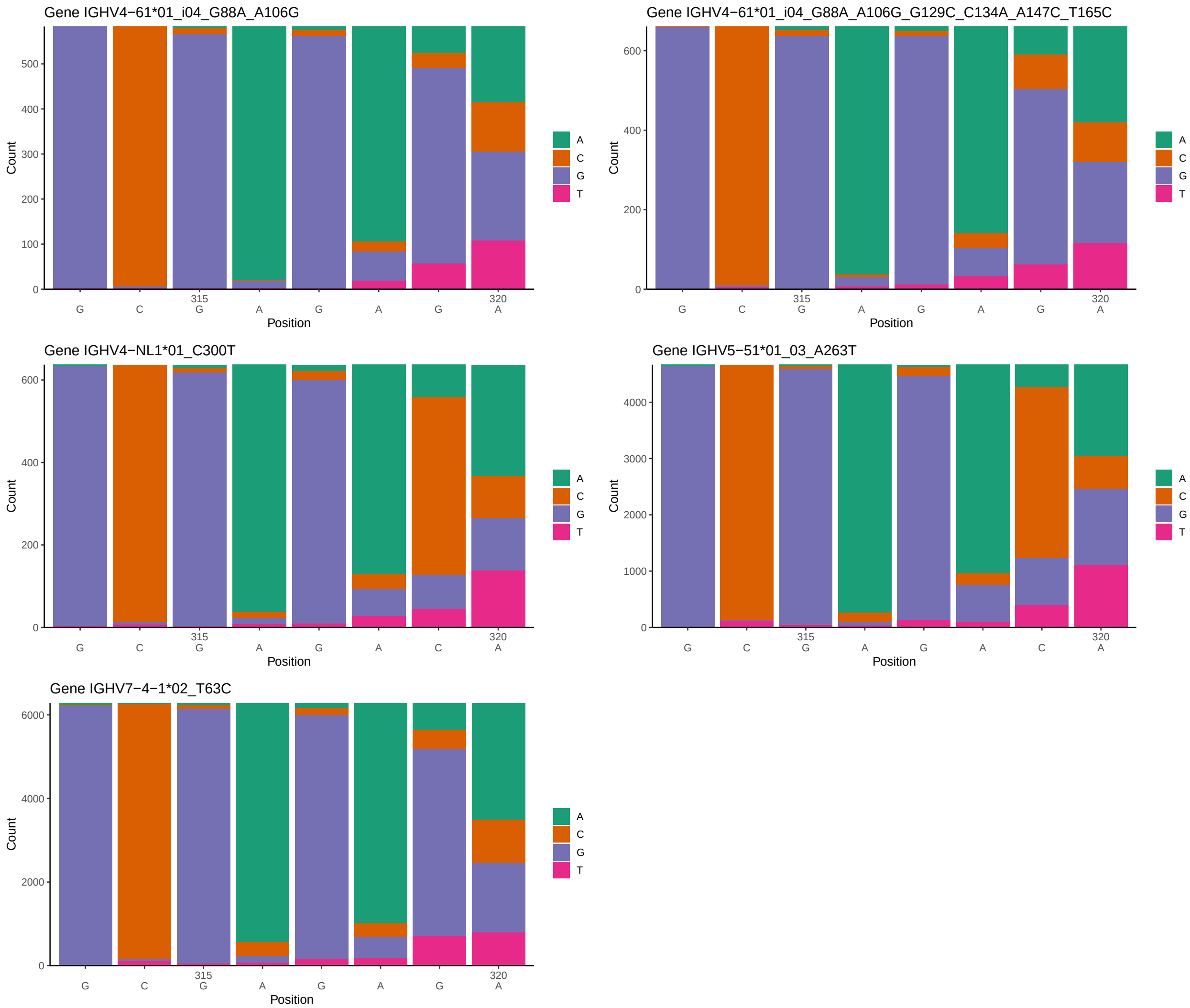
C
G
T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Count
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200
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600
G C 315 G A G A G 320 A
Position
A C G T
Gene IGHV4−NL1*01_C300T
Count
0
200
400
600
G C 315 G A G A C 320 A
Position
A C G T
Gene IGHV5−51*01_03_A263T
Count
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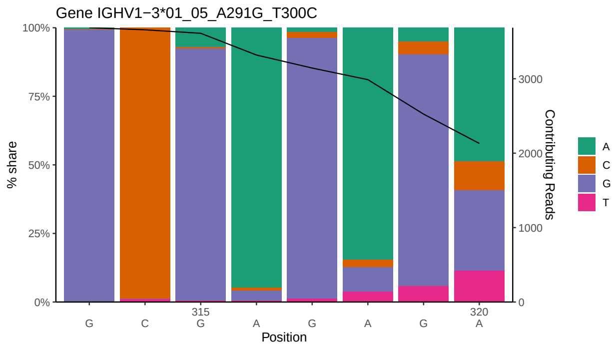
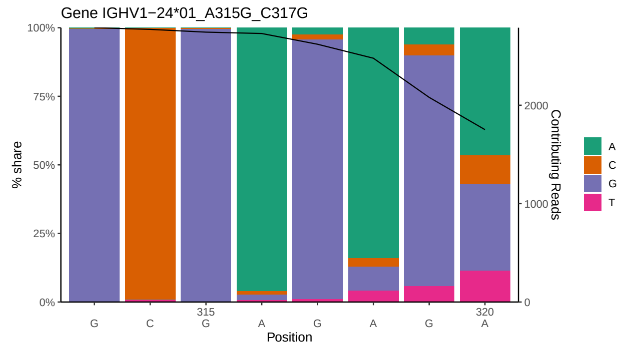
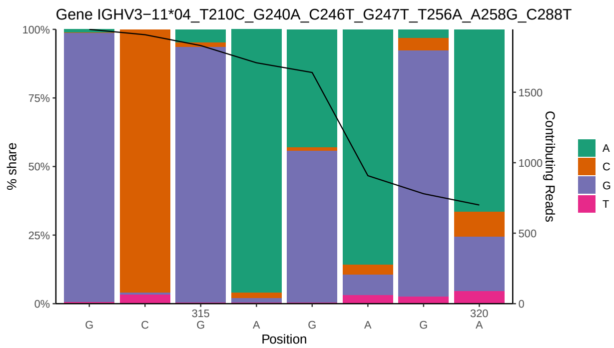
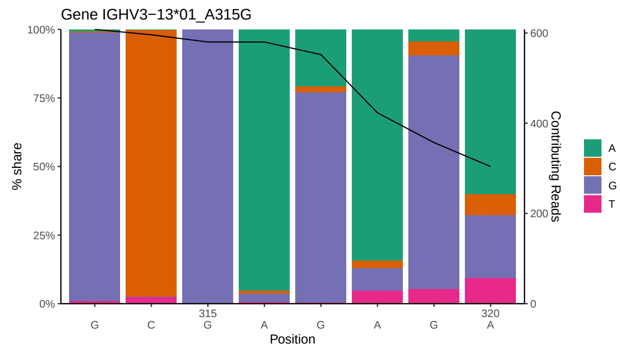
1000
2000
3000
4000
G C 315 G A G A C 320 A
Position
A C G T
Gene IGHV7−4−1*02_T63C
Count
0
2000
4000
6000
G C 315 G A G A G 320 A
Position
A C G T

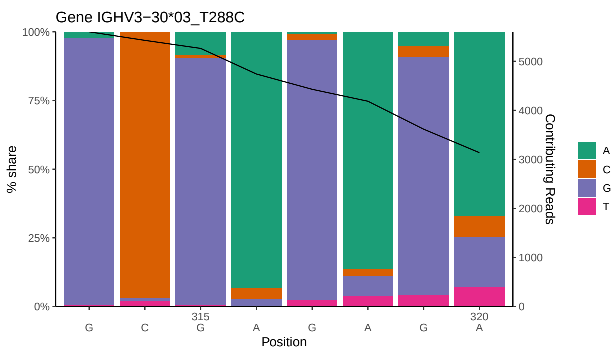
1.2 Per-nucleotide consensus where previous nucleotides match the consensus

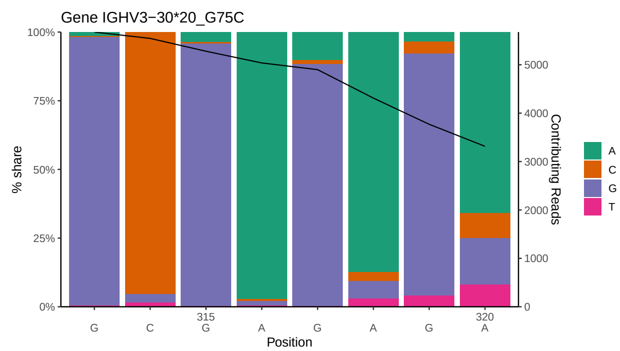


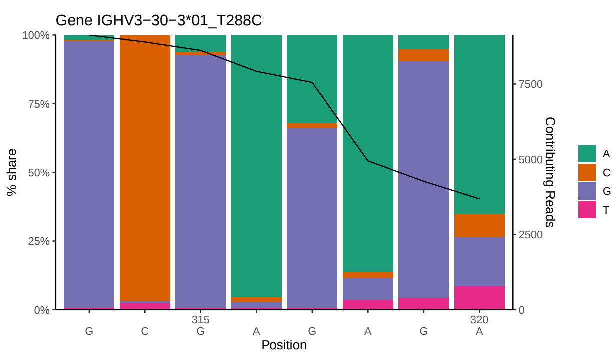


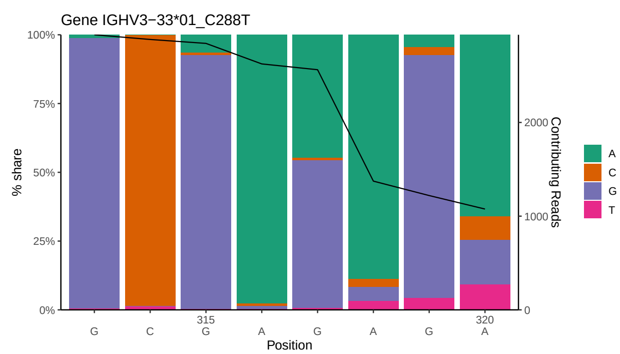


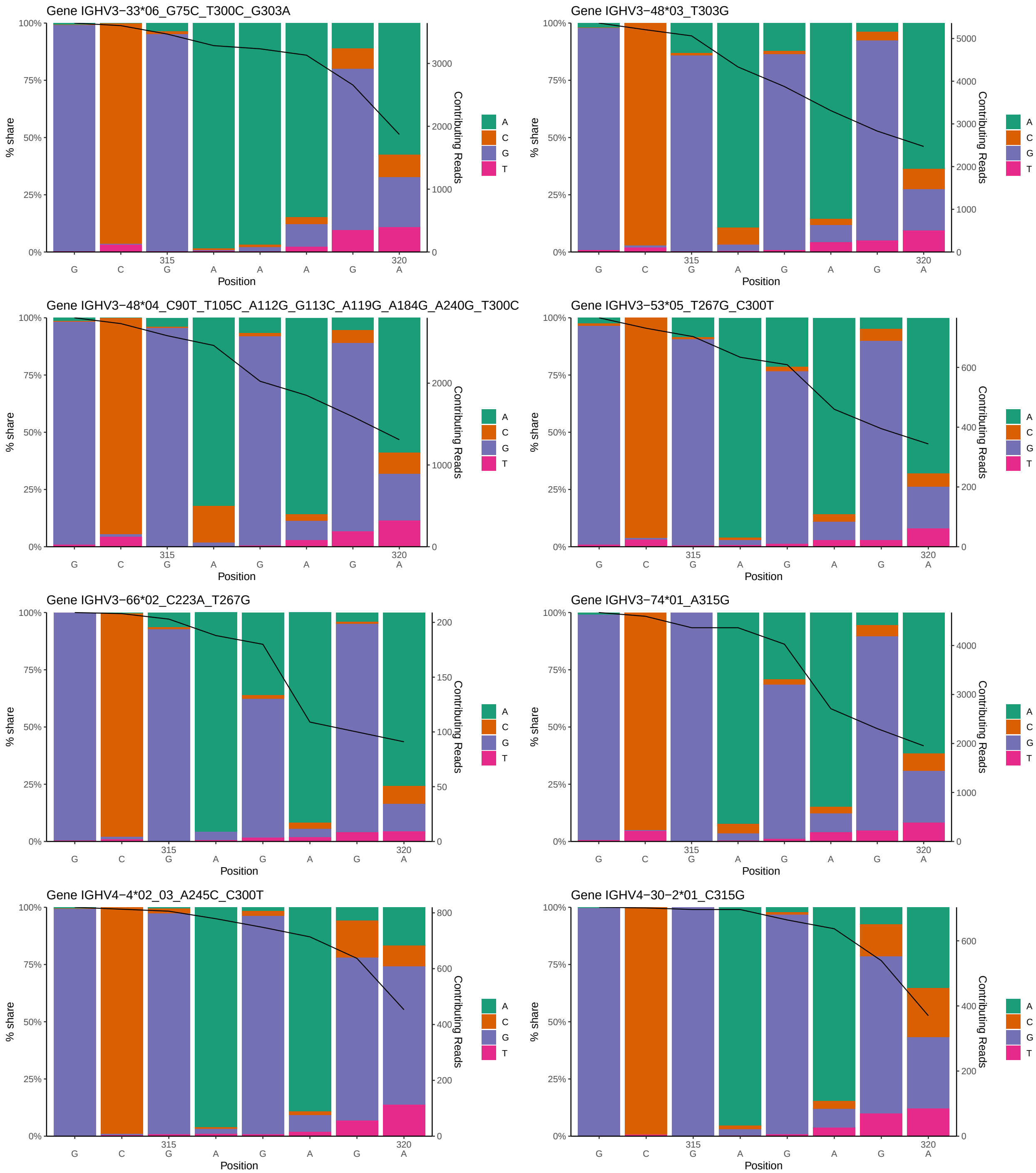


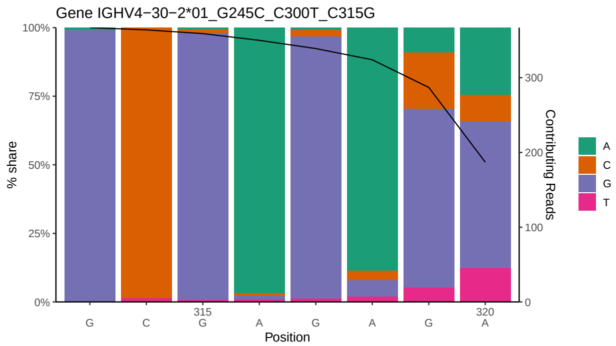
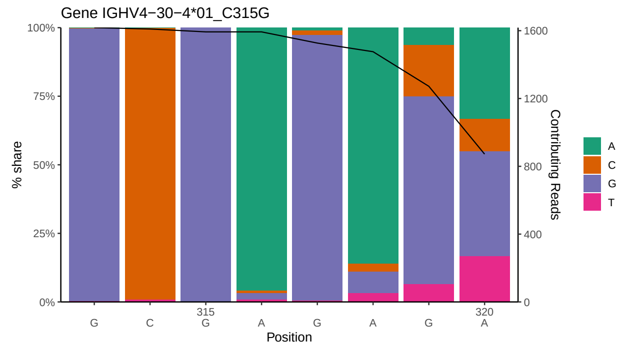









Gene IGHV3−33*06_G75C_T300C_G303A
Gene IGHV3−48*03_T303G
Gene IGHV3−48*04_C90T_T105C_A112G_G113C_A119G_A184G_A240G_T300C
Gene IGHV3−53*05_T267G_C300T
Gene IGHV3−66*02_C223A_T267G
Gene IGHV3−74*01_A315G
Gene IGHV4−4*02_03_A245C_C300T
Gene IGHV4−30−2*01_C315G
% share
Contributing Reads
Position
G C G A A A G A
G C G A G A G A
A
C
G
T


Gene IGHV4−30−2*01_G245C_C300T_C315G
% share
Contributing Reads
Position
A
C
G
T

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Contributing Reads
Position
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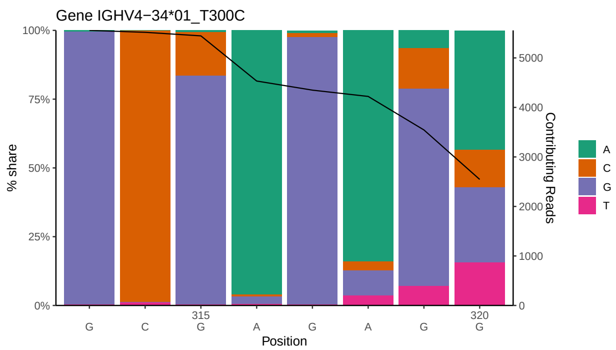
C
G
T

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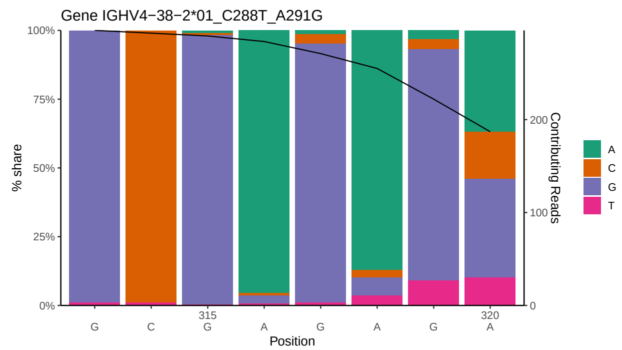
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Contributing Reads
Position
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C
G
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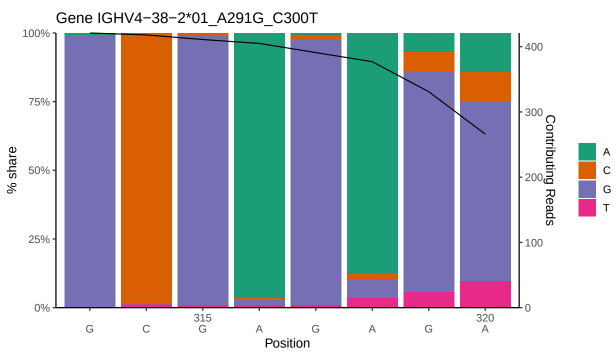
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Contributing Reads
Position
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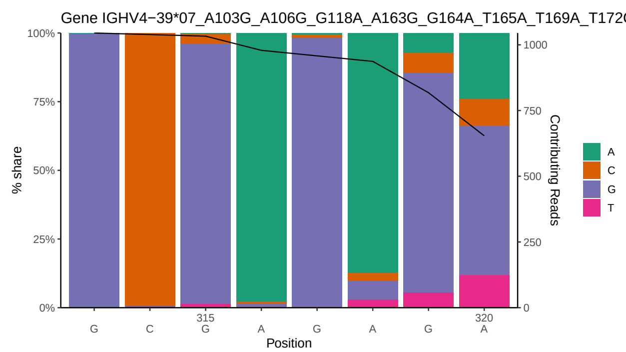
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Contributing Reads
Position
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C
G
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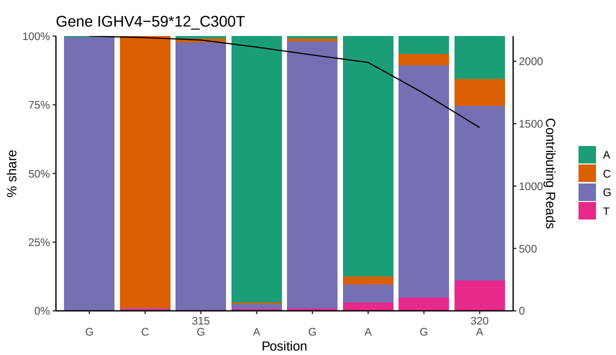
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% share
Contributing Reads
Position
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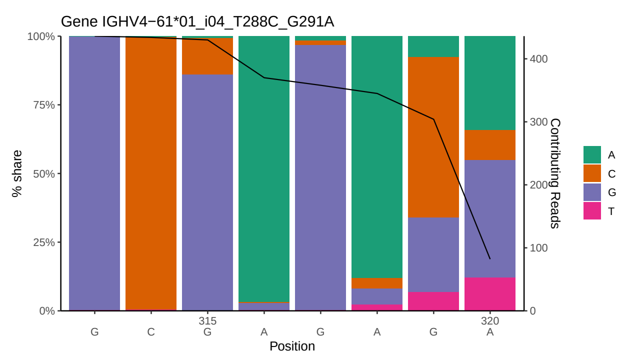
C
G
T

Gene IGHV4−61*01_i04_T288C_G291A
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Contributing Reads
Position
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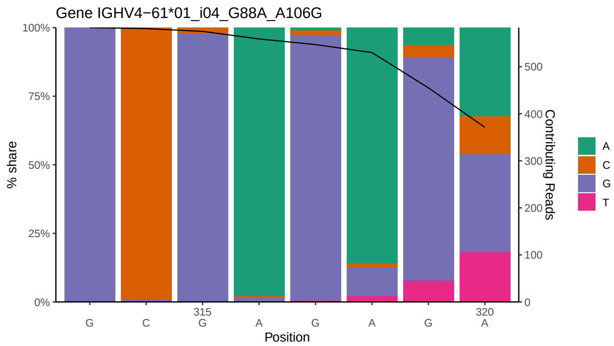
C
G
T


Gene IGHV4−61*01_i04_G88A_A106G
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Contributing Reads
Position
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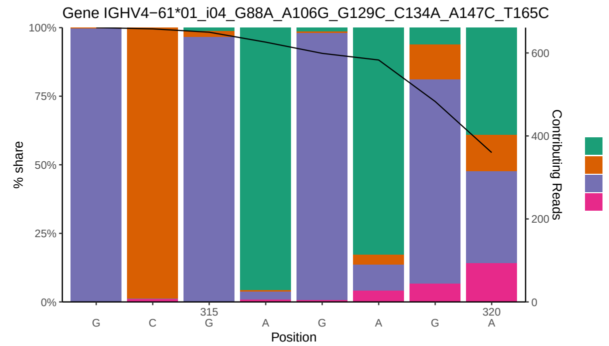
A
C
G
T


Gene IGHV4−61*01_i04_G88A_A106G_G129C_C134A_A147C_T165C
% share
Contributing Reads
Position
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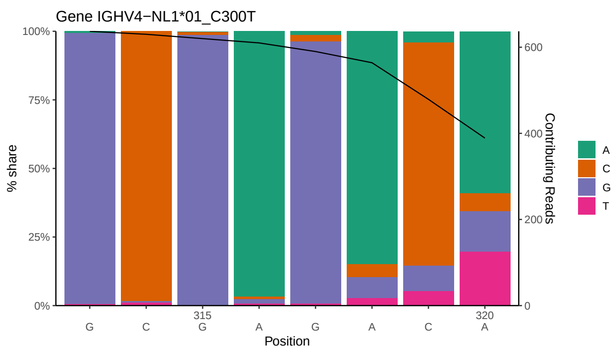
A
C
G
T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C
G
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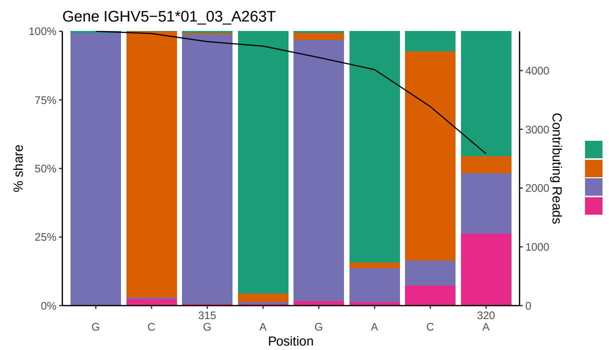
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Contributing Reads
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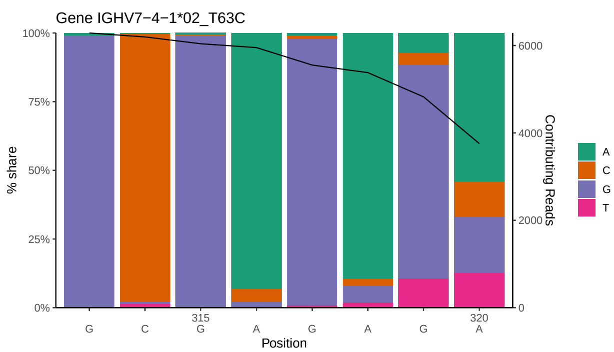
A
C
G
T


Gene IGHV7−4−1*02_T63C
% share
Contributing Reads
Position
G C G A G A G A
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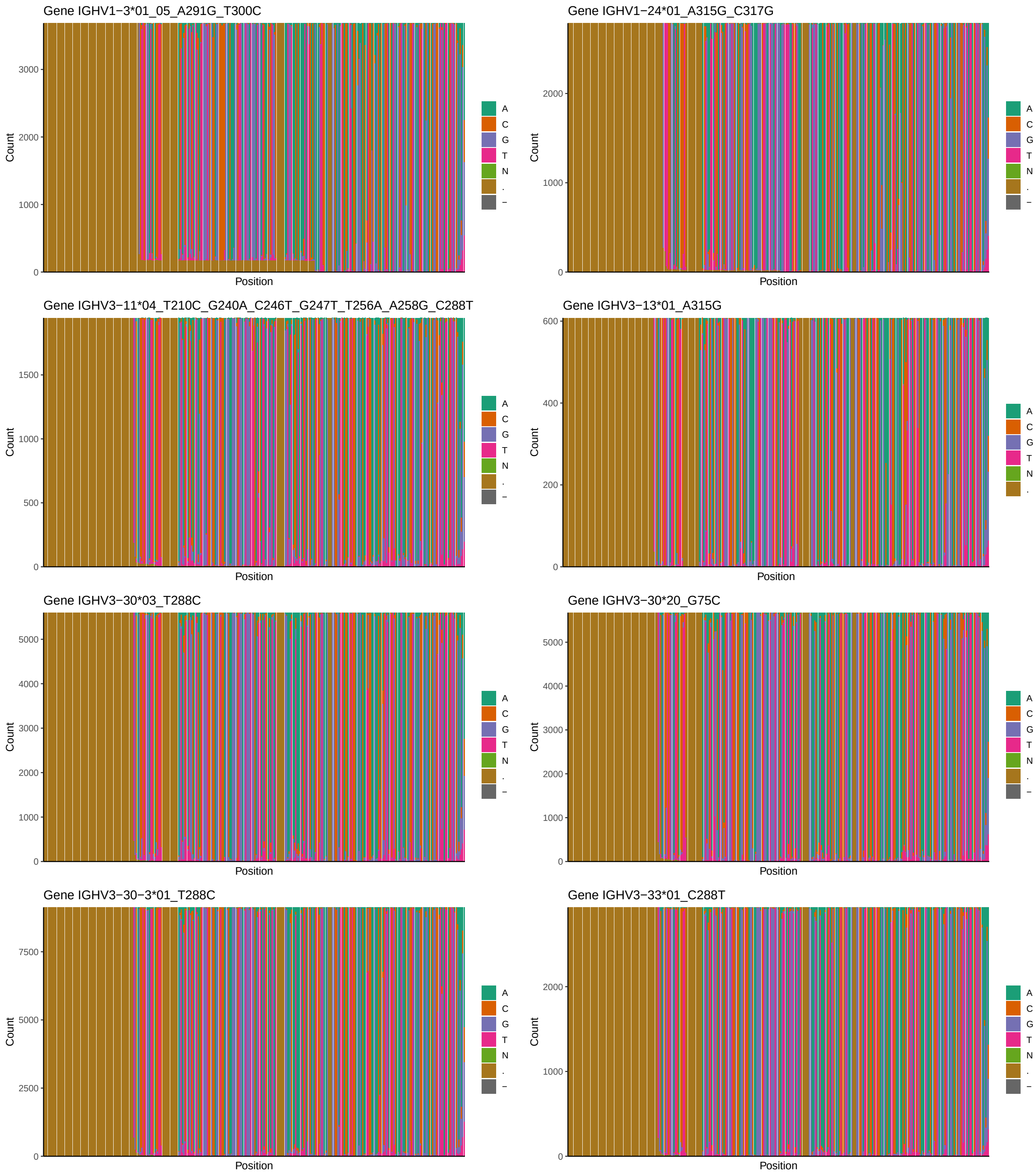
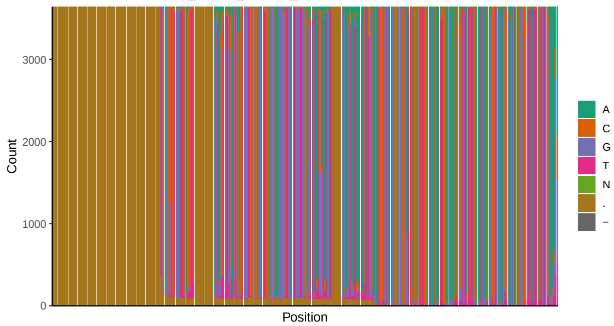
C
G
T

1.3 Whole-sequence composition of each assigned read



Gene IGHV3−33*06_G75C_T300C_G303A



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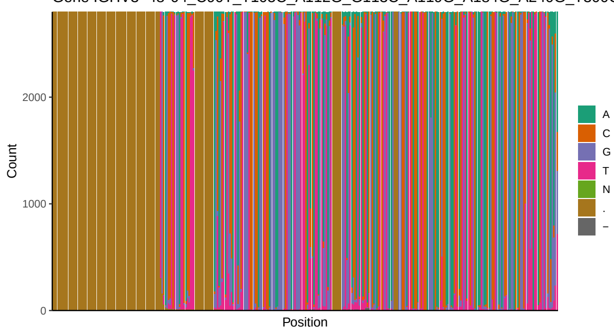
Gene IGHV3−48*04_C90T_T105C_A112G_G113C_A119G_A184G_A240G_T300C



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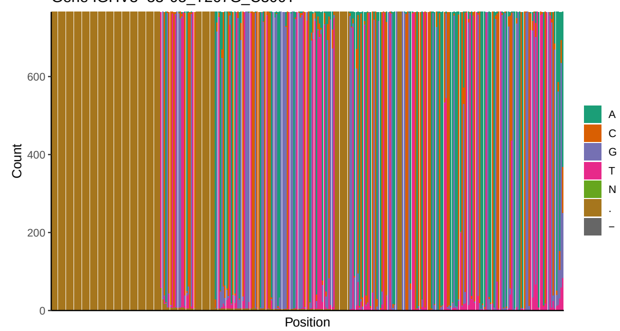

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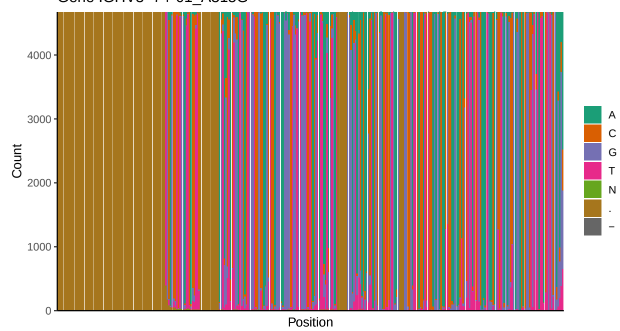
Gene IGHV3−74*01_A315G



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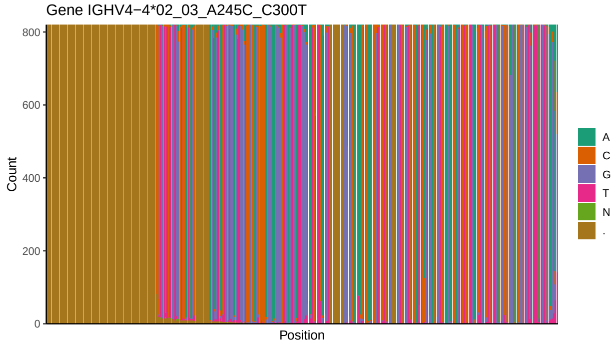

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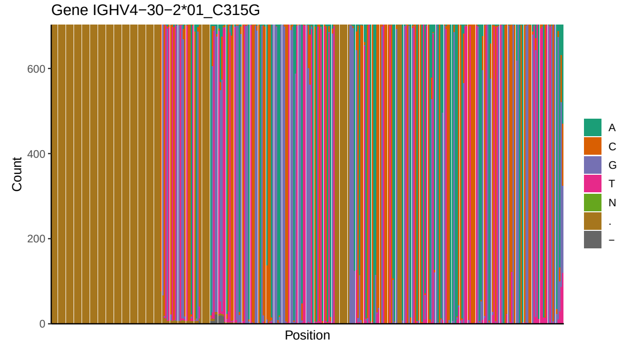

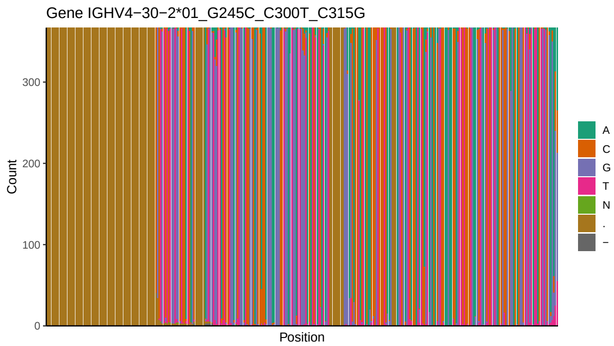
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Position
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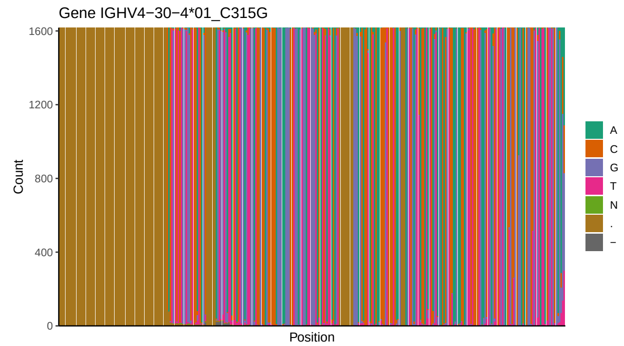
C
G
T
N
.
−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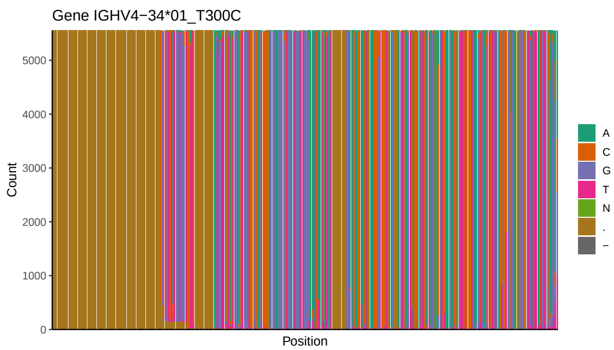
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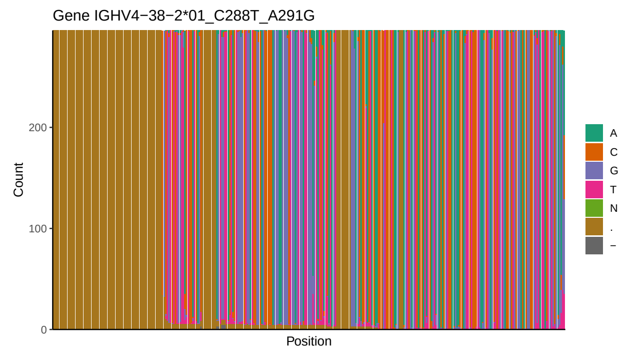
−


Gene IGHV4−38−2*01_C288T_A291G
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Position
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G
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N
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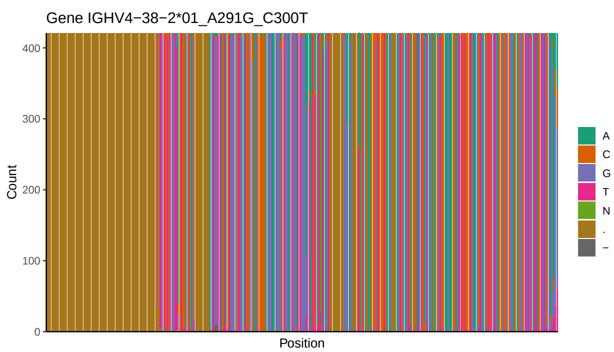
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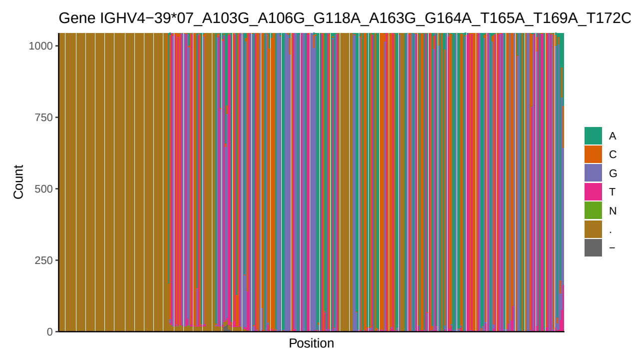
200
300
400
Position
A
C
G
T
N
.
−


Gene IGHV4−39*07_A103G_A106G_G118A_A163G_G164A_T165A_T169A_T172C
Count
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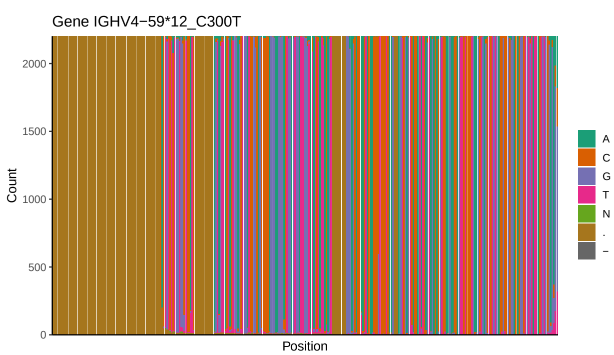
500
750
1000
Position
A
C
G
T
N
.
−


Gene IGHV4−59*12_C300T
Count
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500
1000
1500
2000
Position
A
C
G
T
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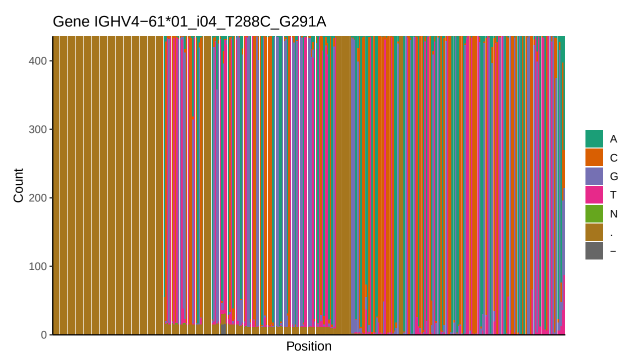
.
−


Gene IGHV4−61*01_i04_T288C_G291A
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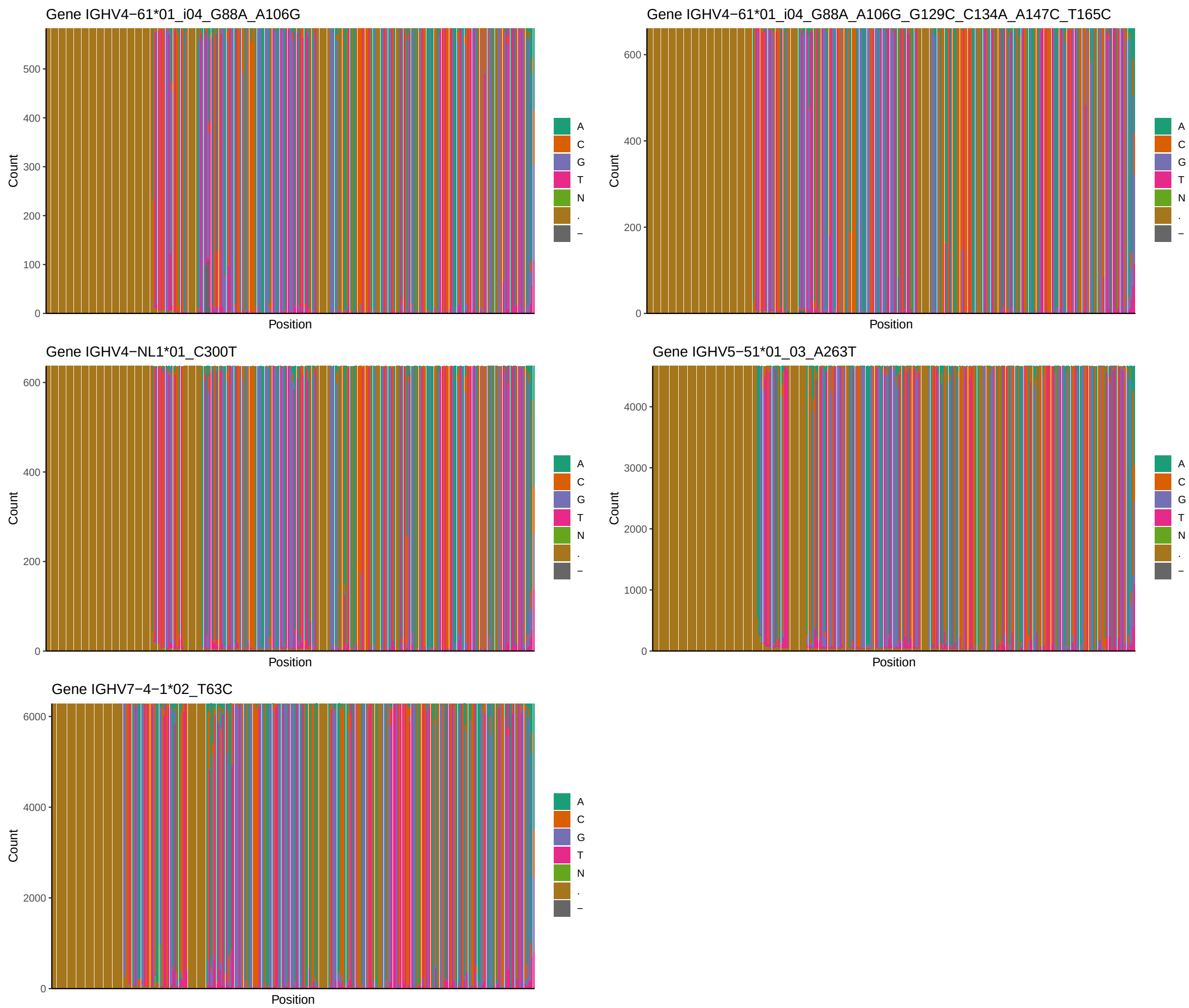
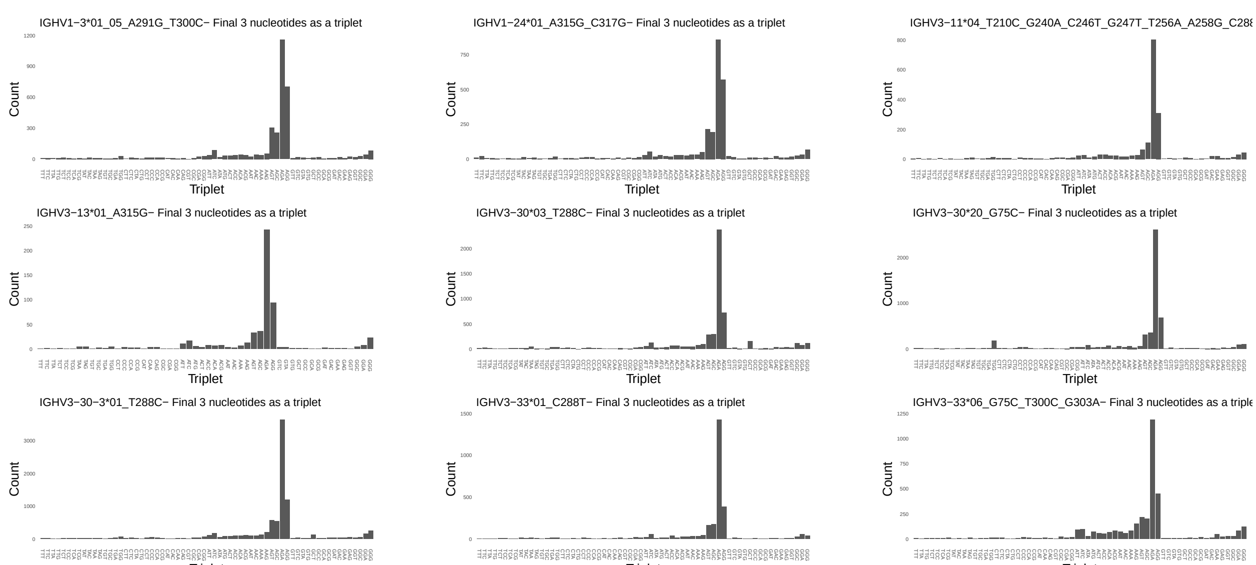
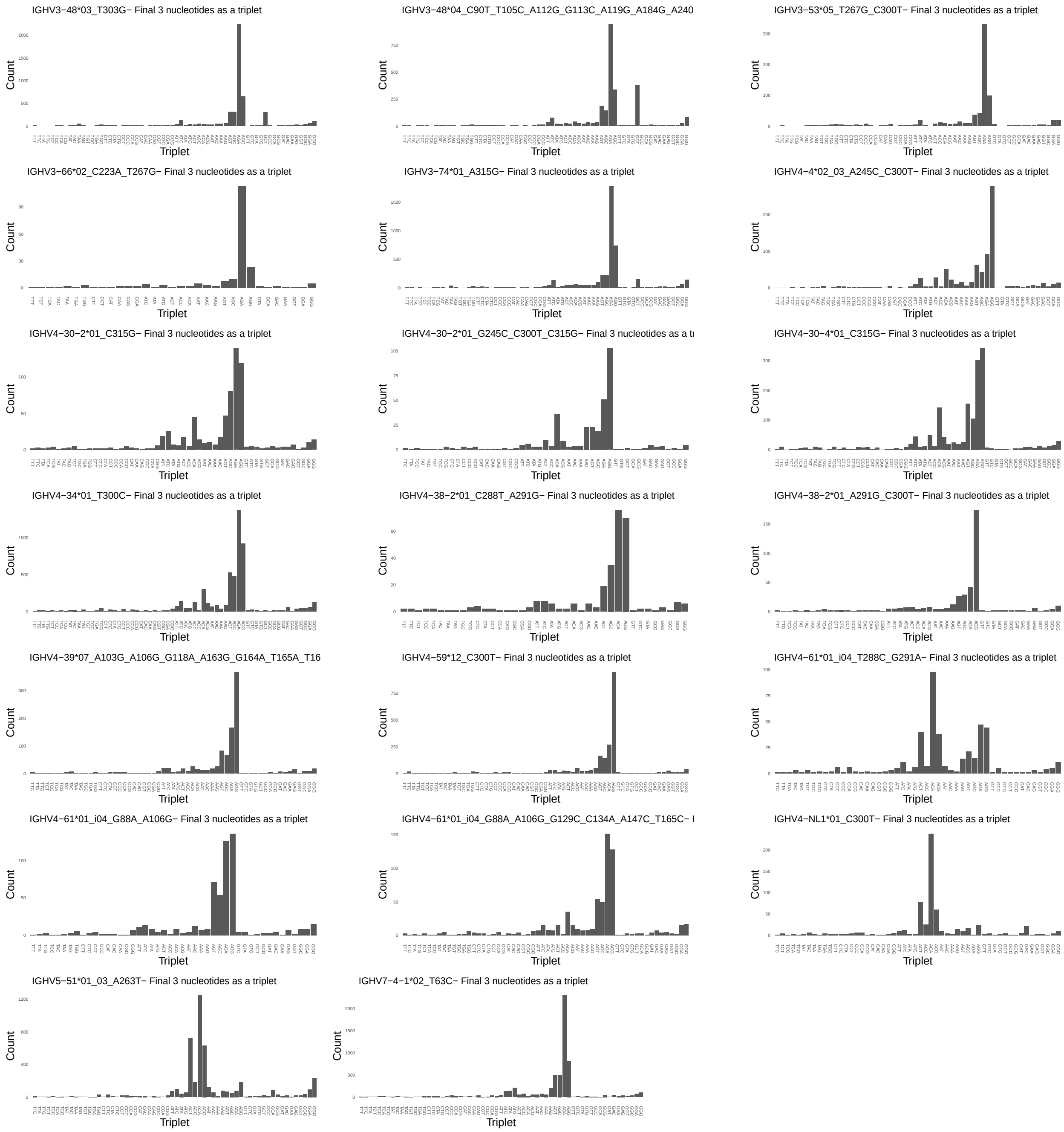
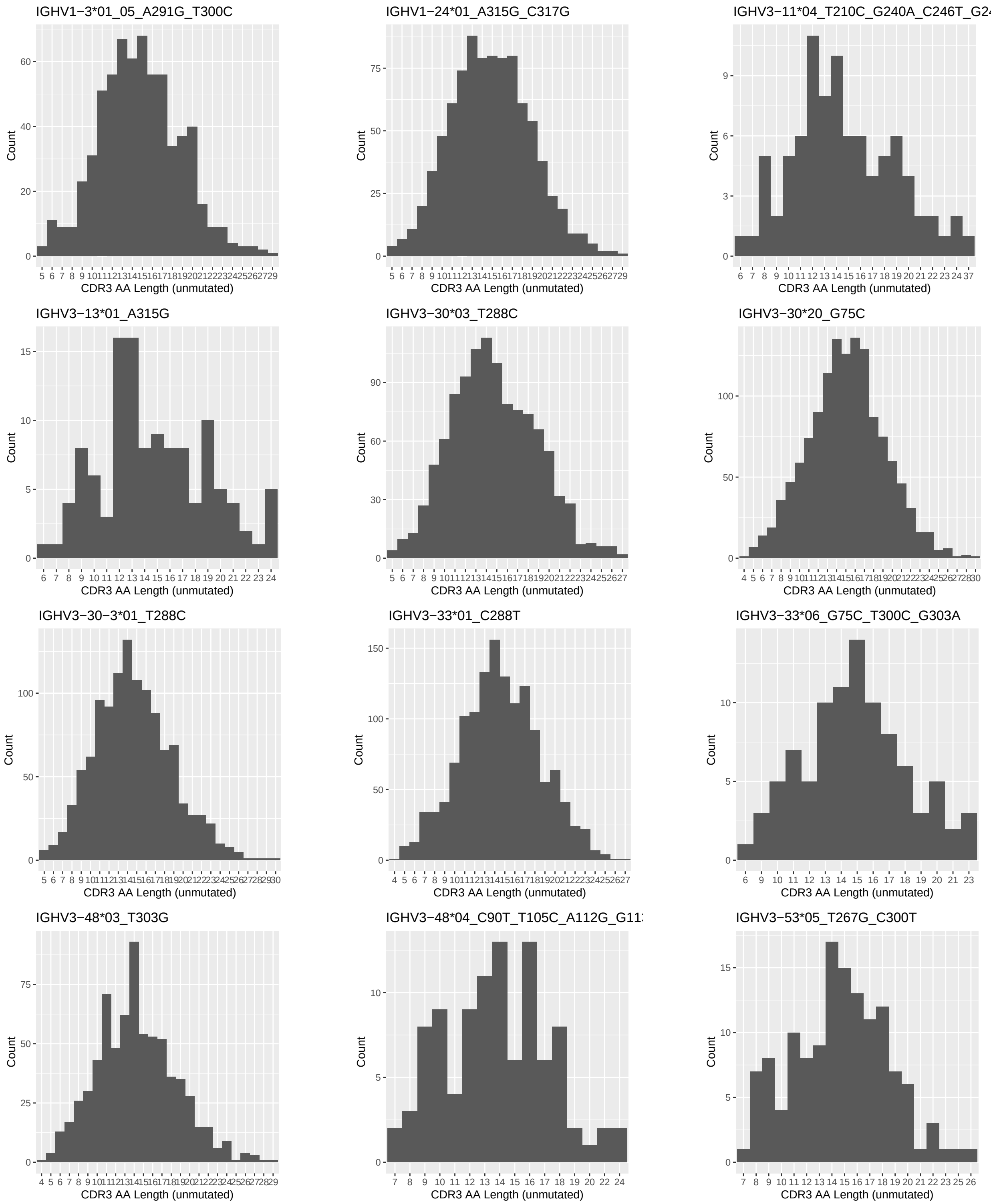


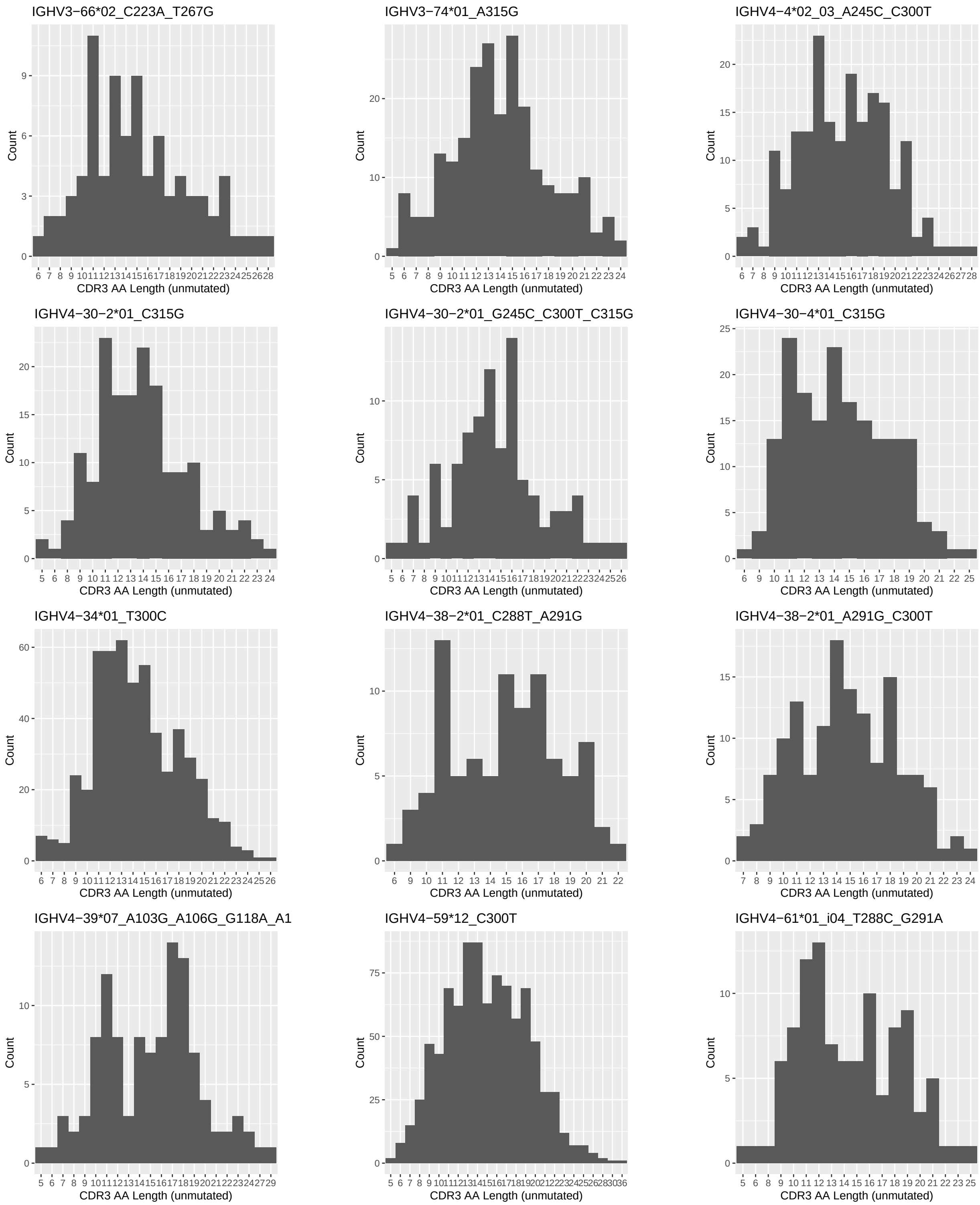
1.4 Final three nucleotides: frequency of each observed triplet

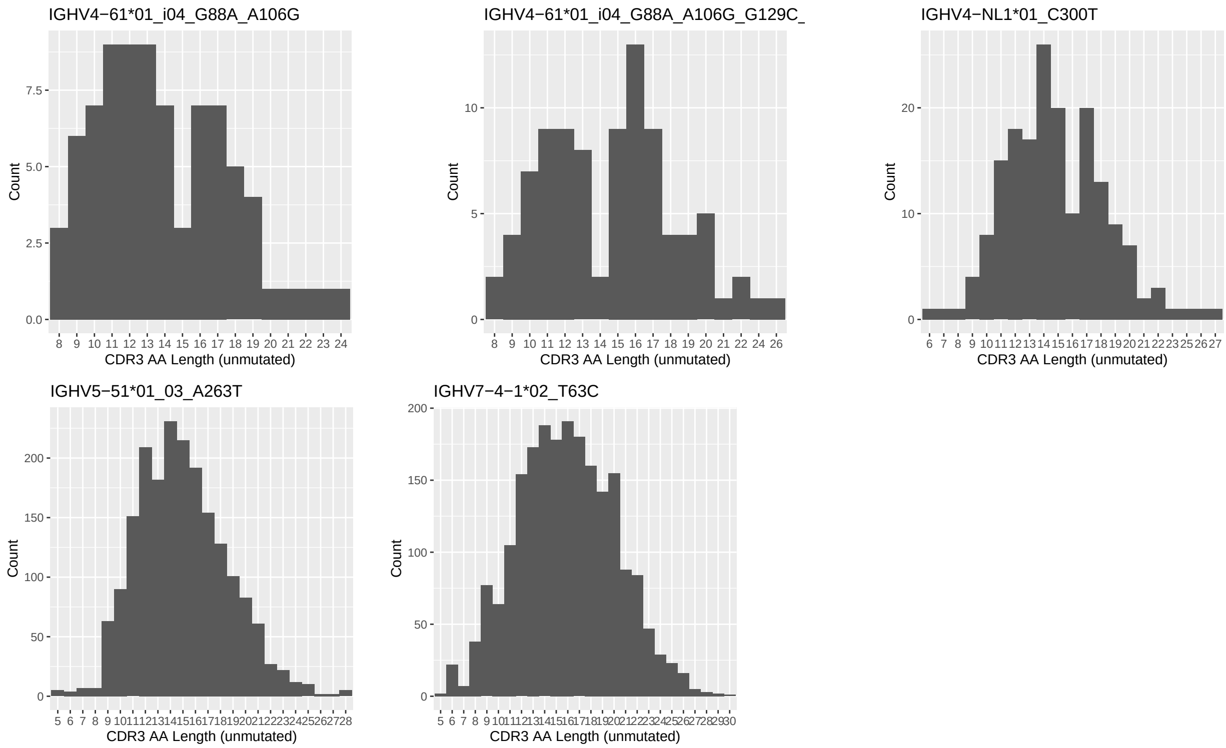



IGHV3−48*03_T303G− Final 3 nucleotides as a triplet
IGHV3−48*04_C90T_T105C_A112G_G113C_A119G_A184G_A240
IGHV3−53*05_T267G_C300T− Final 3 nucleotides as a triplet
IGHV3−66*02_C223A_T267G− Final 3 nucleotides as a triplet
IGHV3−74*01_A315G− Final 3 nucleotides as a triplet
IGHV4−4*02_03_A245C_C300T− Final 3 nucleotides as a triplet
IGHV4−30−2*01_C315G− Final 3 nucleotides as a triplet
IGHV4−30−2*01_G245C_C300T_C315G− Final 3 nucleotides as a t
IGHV4−30−4*01_C315G− Final 3 nucleotides as a triplet
IGHV4−34*01_T300C− Final 3 nucleotides as a triplet
IGHV4−38−2*01_C288T_A291G− Final 3 nucleotides as a triplet
IGHV4−38−2*01_A291G_C300T− Final 3 nucleotides as a triplet
IGHV4−39*07_A103G_A106G_G118A_A163G_G164A_T165A_T16
IGHV4−59*12_C300T− Final 3 nucleotides as a triplet
IGHV4−61*01_i04_T288C_G291A− Final 3 nucleotides as a triplet
IGHV4−61*01_i04_G88A_A106G− Final 3 nucleotides as a triplet
IGHV4−61*01_i04_G88A_A106G_G129C_C134A_A147C_T165C−
IGHV4−NL1*01_C300T− Final 3 nucleotides as a triplet
IGHV5−51*01_03_A263T− Final 3 nucleotides as a triplet
IGHV7−4−1*02_T63C− Final 3 nucleotides as a triplet
Count
Triplet

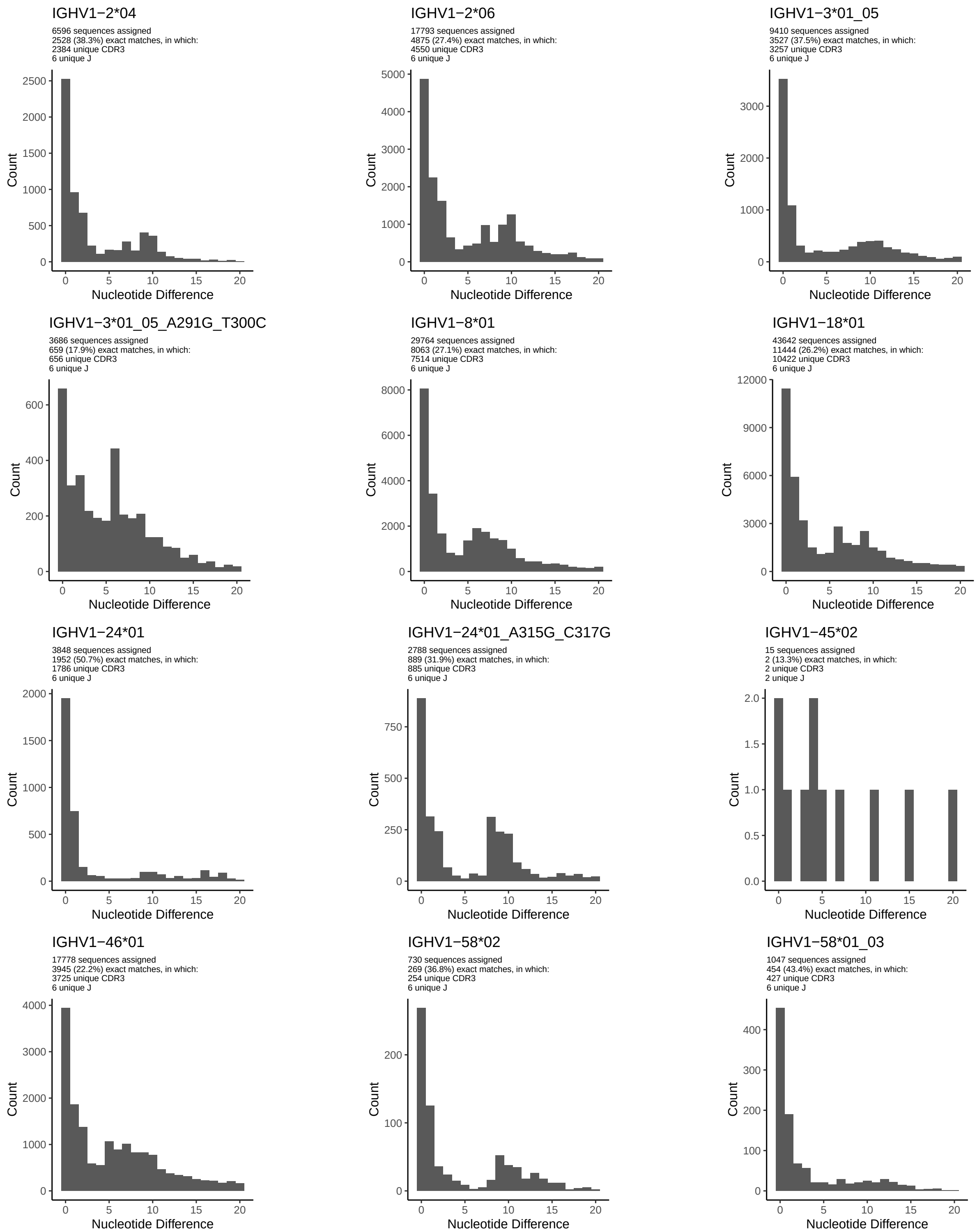
1.5 CDR3 length distribution, in assignments to novel alleles

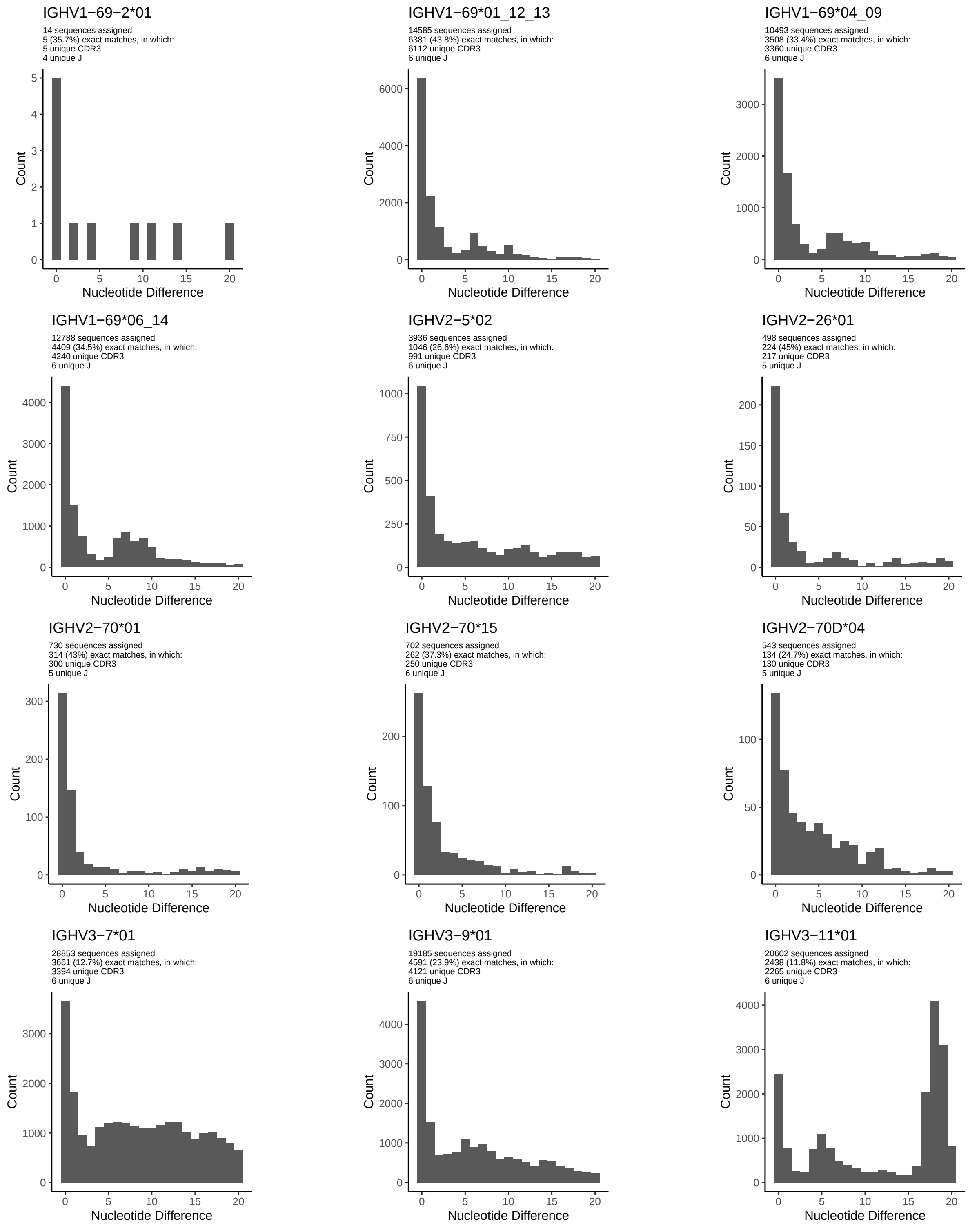


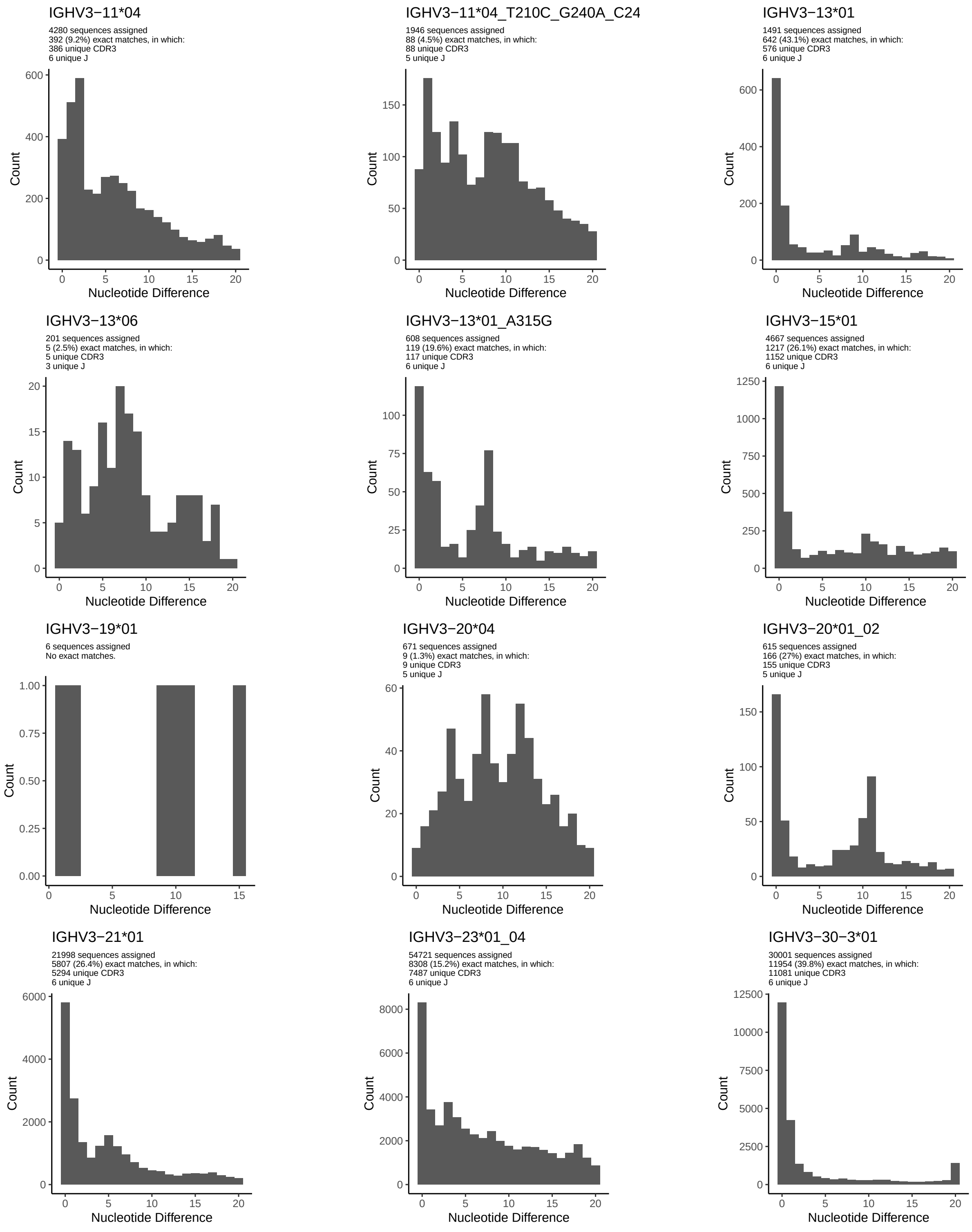

IGHV3−66*02_C223A_T267G
IGHV3−74*01_A315G
IGHV4−4*02_03_A245C_C300T
IGHV4−30−2*01_C315G
IGHV4−30−2*01_G245C_C300T_C315G
IGHV4−30−4*01_C315G
IGHV4−34*01_T300C
IGHV4−38−2*01_C288T_A291G
IGHV4−38−2*01_A291G_C300T
IGHV4−39*07_A103G_A106G_G118A_A1
IGHV4−59*12_C300T
IGHV4−61*01_i04_T288C_G291A
Count
CDR3 AA Length (unmutated)

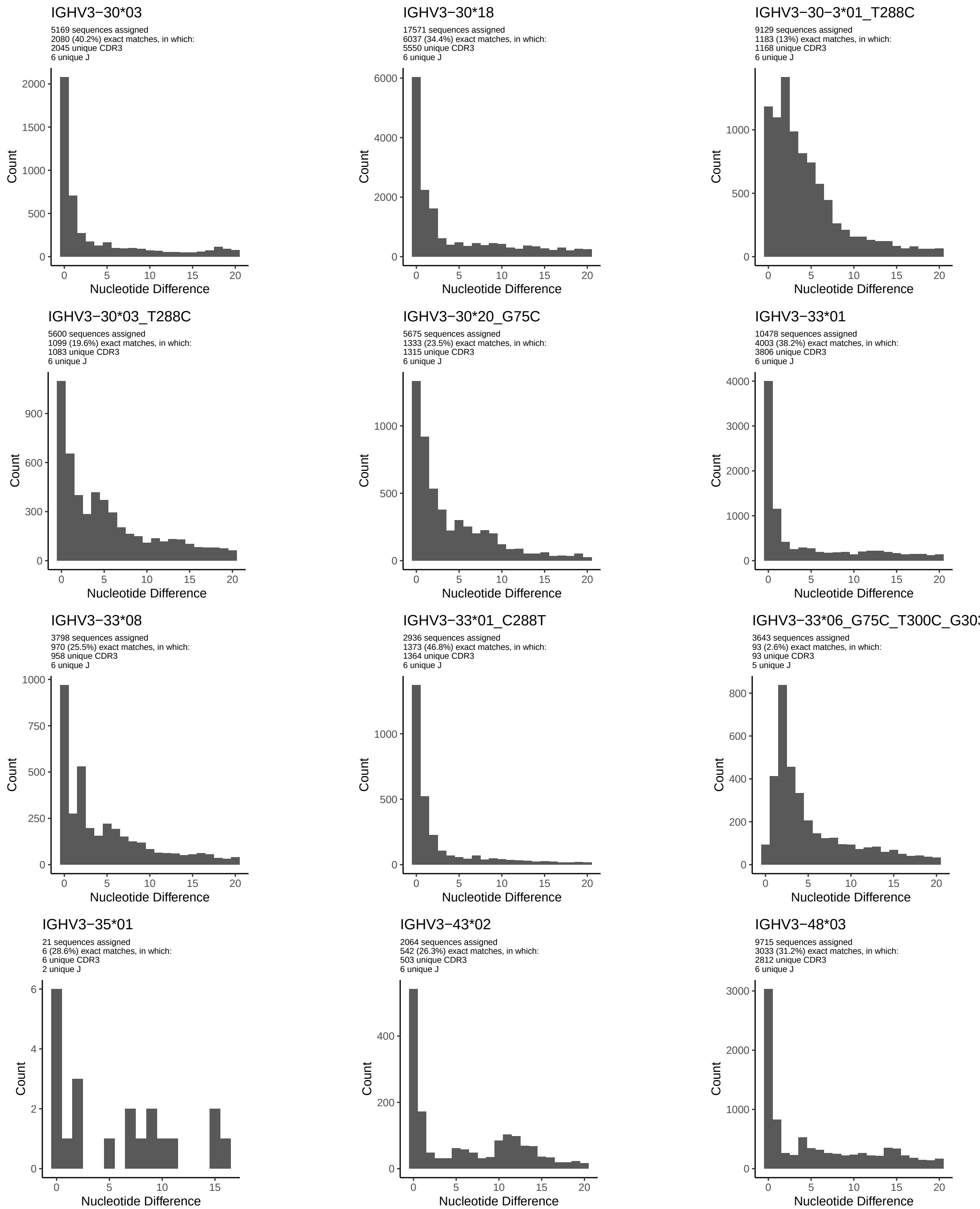

IGHV4−61*01_i04_G88A_A106G
IGHV4−61*01_i04_G88A_A106G_G129C_
IGHV4−NL1*01_C300T
IGHV5−51*01_03_A263T
IGHV7−4−1*02_T63C
Count
CDR3 AA Length (unmutated)

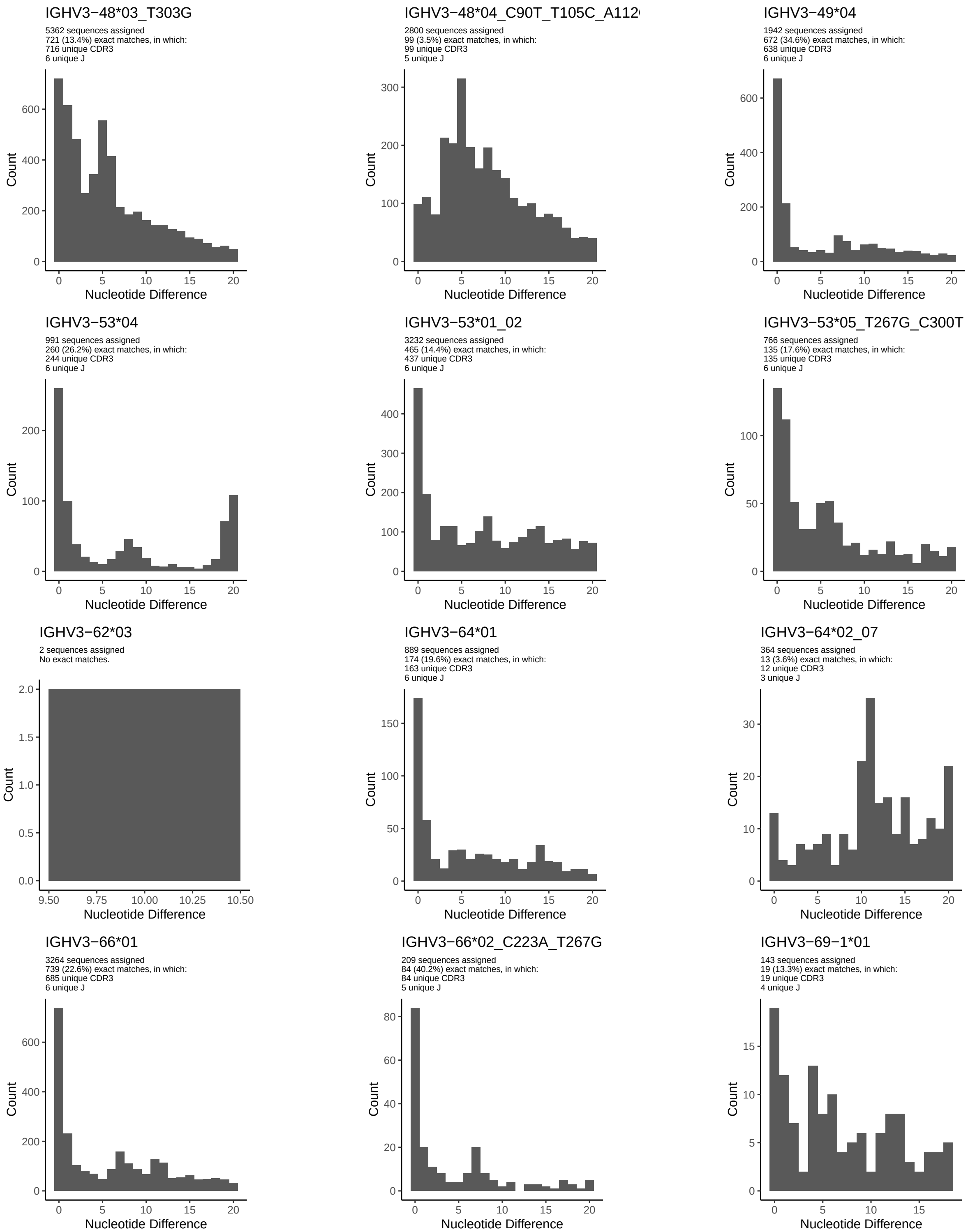
2 Variation from germline, in assignments to each allele



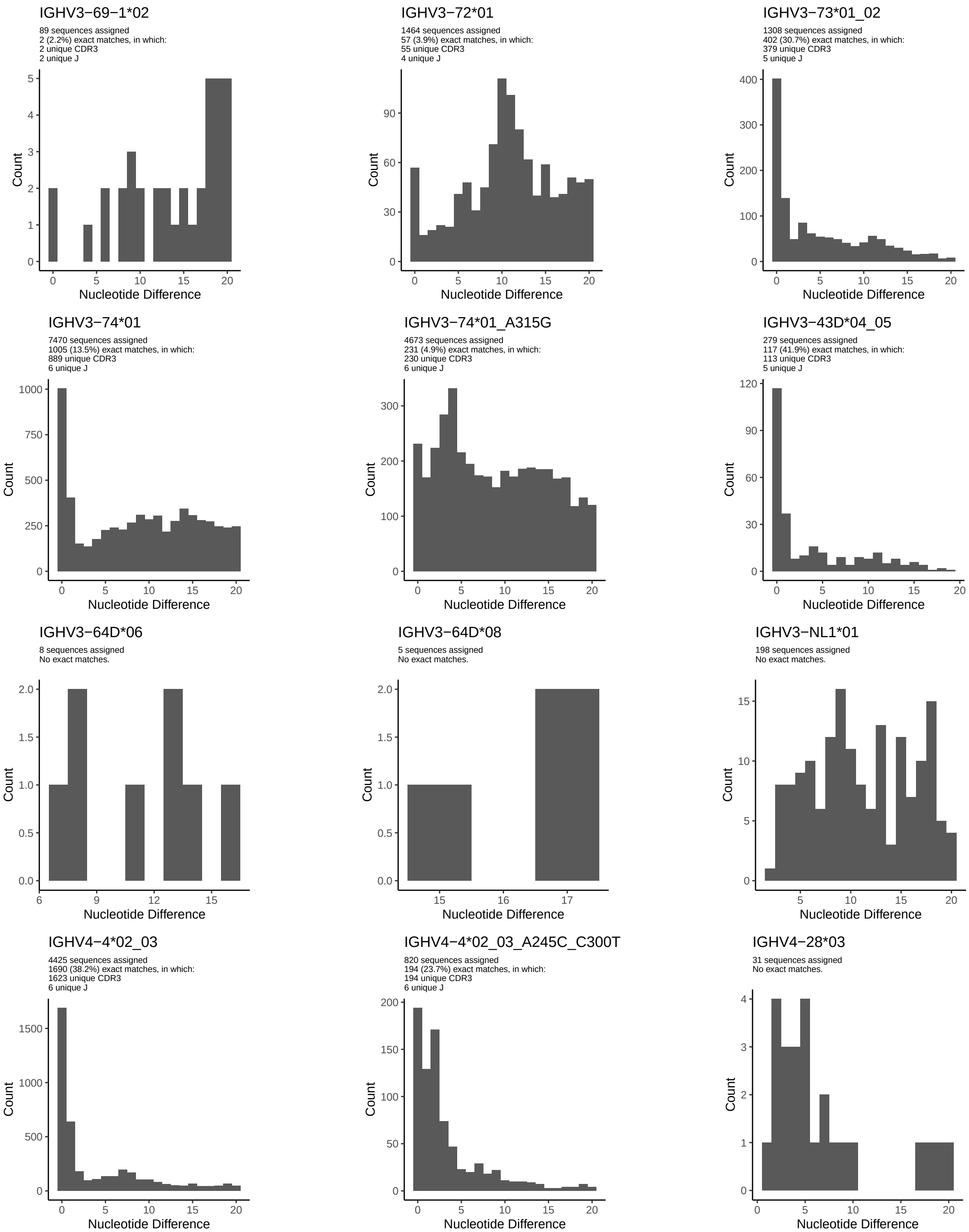

IGHV1−69−2*01
14 sequences assigned
5 (35.7%) exact matches, in which:
5 unique CDR3
4 unique J
IGHV1−69*01_12_13
14585 sequences assigned
6381 (43.8%) exact matches, in which:
6112 unique CDR3
6 unique J
IGHV1−69*04_09
10493 sequences assigned
3508 (33.4%) exact matches, in which:
3360 unique CDR3
6 unique J
IGHV1−69*06_14
12788 sequences assigned
4409 (34.5%) exact matches, in which:
4240 unique CDR3
6 unique J
IGHV2−5*02
3936 sequences assigned
1046 (26.6%) exact matches, in which:
991 unique CDR3
6 unique J
IGHV2−26*01
498 sequences assigned
224 (45%) exact matches, in which:
217 unique CDR3
5 unique J
IGHV2−70*01
730 sequences assigned
314 (43%) exact matches, in which:
300 unique CDR3
5 unique J
IGHV2−70*15
702 sequences assigned
262 (37.3%) exact matches, in which:
250 unique CDR3
6 unique J
IGHV2−70D*04
543 sequences assigned
134 (24.7%) exact matches, in which:
130 unique CDR3
5 unique J
IGHV3−7*01
28853 sequences assigned
3661 (12.7%) exact matches, in which:
3394 unique CDR3
6 unique J
IGHV3−9*01
19185 sequences assigned
4591 (23.9%) exact matches, in which:
4121 unique CDR3
6 unique J
IGHV3−11*01
20602 sequences assigned
2438 (11.8%) exact matches, in which:
2265 unique CDR3
6 unique J
Count
Nucleotide Difference

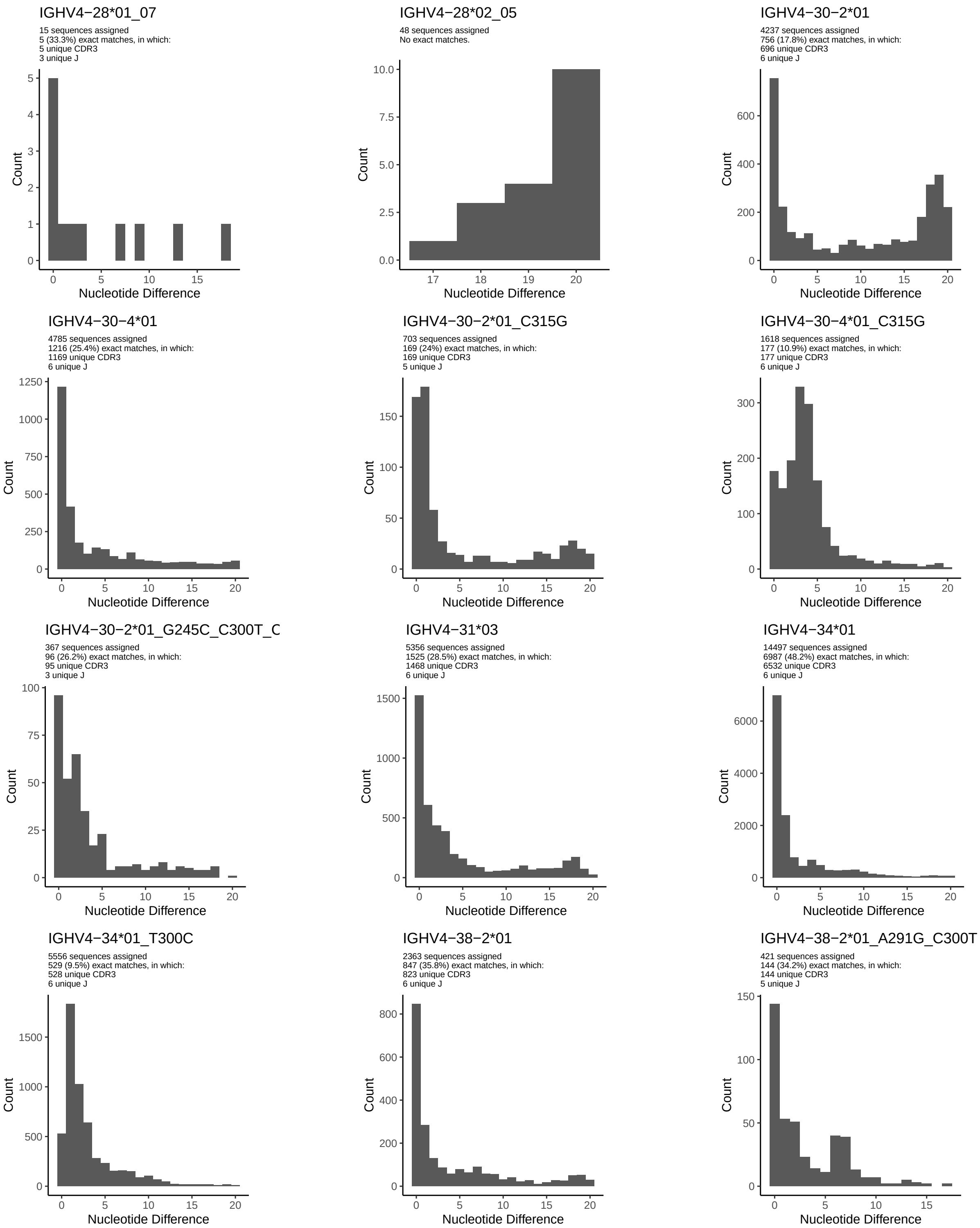

IGHV3−11*04
4280 sequences assigned
392 (9.2%) exact matches, in which:
386 unique CDR3
6 unique J
IGHV3−11*04_T210C_G240A_C24
1946 sequences assigned
88 (4.5%) exact matches, in which:
88 unique CDR3
5 unique J
IGHV3−13*01
1491 sequences assigned
642 (43.1%) exact matches, in which:
576 unique CDR3
6 unique J
IGHV3−13*06
201 sequences assigned
5 (2.5%) exact matches, in which:
5 unique CDR3
3 unique J
IGHV3−13*01_A315G
608 sequences assigned
119 (19.6%) exact matches, in which:
117 unique CDR3
6 unique J
IGHV3−15*01
4667 sequences assigned
1217 (26.1%) exact matches, in which:
1152 unique CDR3
6 unique J
IGHV3−19*01
6 sequences assigned
No exact matches.
IGHV3−20*04
671 sequences assigned
9 (1.3%) exact matches, in which:
9 unique CDR3
5 unique J
IGHV3−20*01_02
615 sequences assigned
166 (27%) exact matches, in which:
155 unique CDR3
5 unique J
IGHV3−21*01
21998 sequences assigned
5807 (26.4%) exact matches, in which:
5294 unique CDR3
6 unique J
IGHV3−23*01_04
54721 sequences assigned
8308 (15.2%) exact matches, in which:
7487 unique CDR3
6 unique J
IGHV3−30−3*01
30001 sequences assigned
11954 (39.8%) exact matches, in which:
11081 unique CDR3
6 unique J
Count
Nucleotide Difference

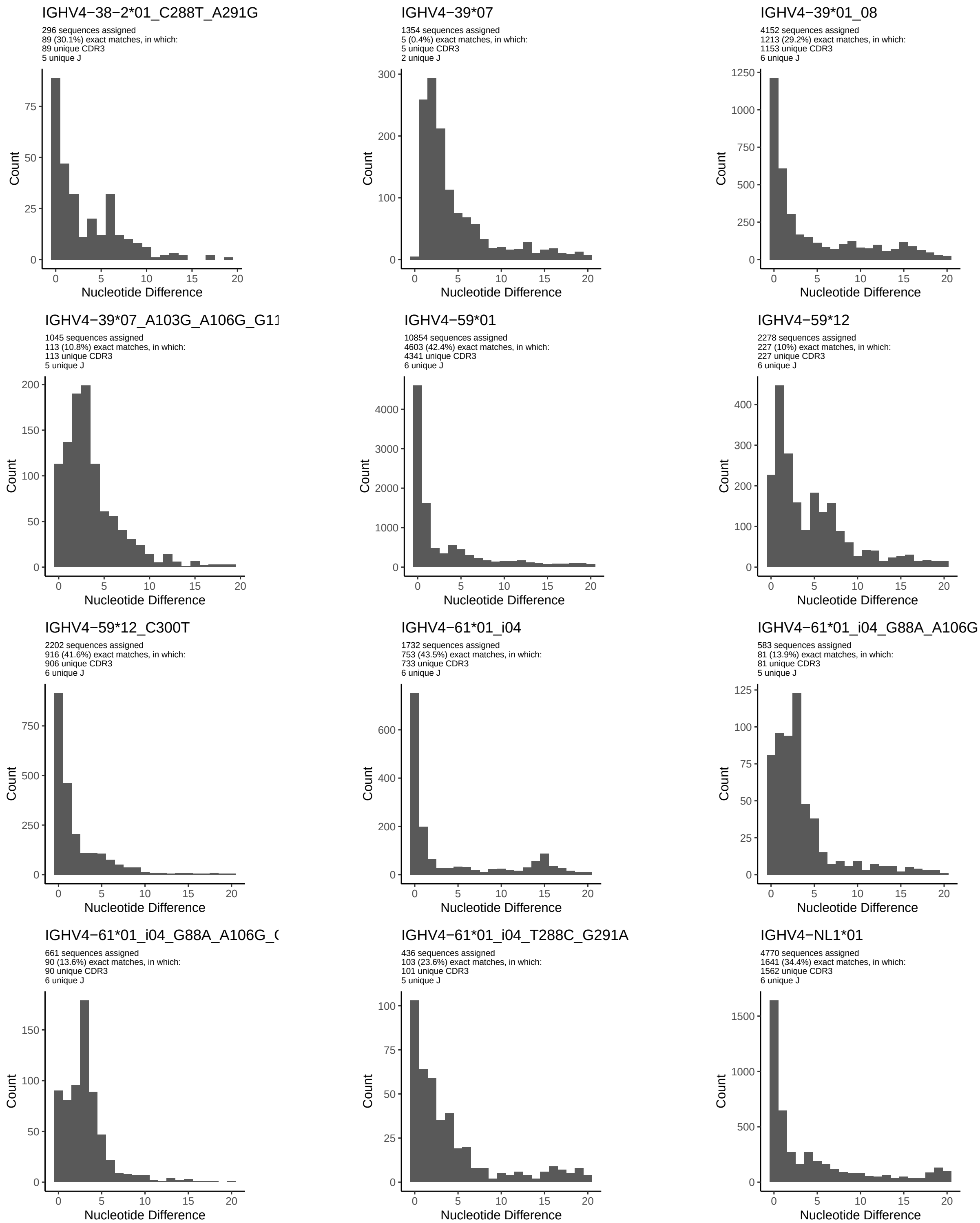

IGHV3−30*03
5169 sequences assigned
2080 (40.2%) exact matches, in which:
2045 unique CDR3
6 unique J
IGHV3−30*18
17571 sequences assigned
6037 (34.4%) exact matches, in which:
5550 unique CDR3
6 unique J
IGHV3−30−3*01_T288C
9129 sequences assigned
1183 (13%) exact matches, in which:
1168 unique CDR3
6 unique J
IGHV3−30*03_T288C
5600 sequences assigned
1099 (19.6%) exact matches, in which:
1083 unique CDR3
6 unique J
IGHV3−30*20_G75C
5675 sequences assigned
1333 (23.5%) exact matches, in which:
1315 unique CDR3
6 unique J
IGHV3−33*01
10478 sequences assigned
4003 (38.2%) exact matches, in which:
3806 unique CDR3
6 unique J
IGHV3−33*08
3798 sequences assigned
970 (25.5%) exact matches, in which:
958 unique CDR3
6 unique J
IGHV3−33*01_C288T
2936 sequences assigned
1373 (46.8%) exact matches, in which:
1364 unique CDR3
6 unique J
IGHV3−33*06_G75C_T300C_G30
3643 sequences assigned
93 (2.6%) exact matches, in which:
93 unique CDR3
5 unique J
IGHV3−35*01
21 sequences assigned
6 (28.6%) exact matches, in which:
6 unique CDR3
2 unique J
IGHV3−43*02
2064 sequences assigned
542 (26.3%) exact matches, in which:
503 unique CDR3
6 unique J
IGHV3−48*03
9715 sequences assigned
3033 (31.2%) exact matches, in which:
2812 unique CDR3
6 unique J
Count
Nucleotide Difference

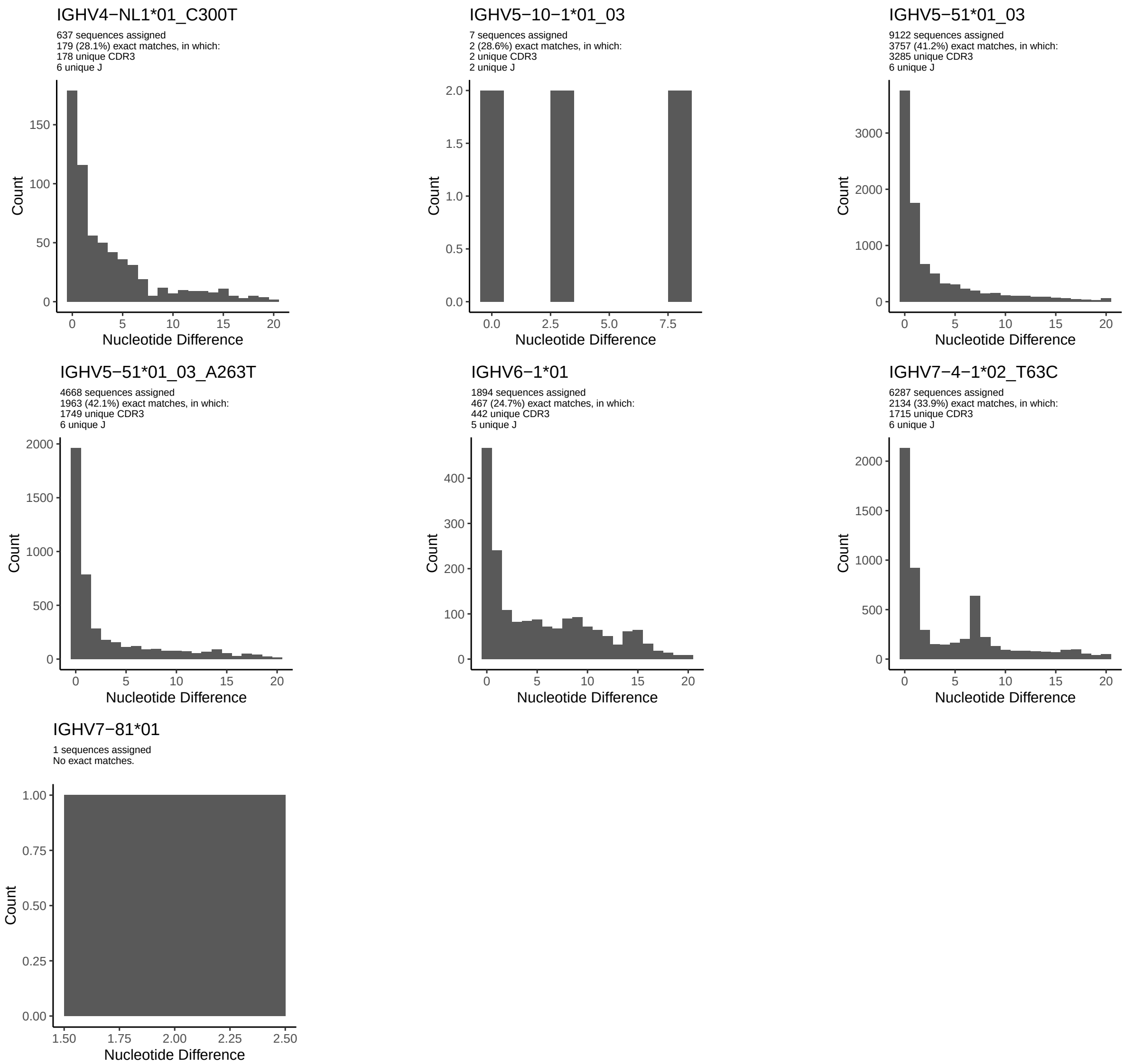


IGHV3−48*03_T303G
5362 sequences assigned
721 (13.4%) exact matches, in which:
716 unique CDR3
6 unique J
IGHV3−48*04_C90T_T105C_A112
2800 sequences assigned
99 (3.5%) exact matches, in which:
99 unique CDR3
5 unique J
IGHV3−49*04
1942 sequences assigned
672 (34.6%) exact matches, in which:
638 unique CDR3
6 unique J
IGHV3−53*04
991 sequences assigned
260 (26.2%) exact matches, in which:
244 unique CDR3
6 unique J
IGHV3−53*01_02
3232 sequences assigned
465 (14.4%) exact matches, in which:
437 unique CDR3
6 unique J
IGHV3−53*05_T267G_C300T
766 sequences assigned
135 (17.6%) exact matches, in which:
135 unique CDR3
6 unique J
IGHV3−62*03
2 sequences assigned
No exact matches.
IGHV3−64*01
889 sequences assigned
174 (19.6%) exact matches, in which:
163 unique CDR3
6 unique J
IGHV3−64*02_07
364 sequences assigned
13 (3.6%) exact matches, in which:
12 unique CDR3
3 unique J
IGHV3−66*01
3264 sequences assigned
739 (22.6%) exact matches, in which:
685 unique CDR3
6 unique J
IGHV3−66*02_C223A_T267G
209 sequences assigned
84 (40.2%) exact matches, in which:
84 unique CDR3
5 unique J
IGHV3−69−1*01
143 sequences assigned
19 (13.3%) exact matches, in which:
19 unique CDR3
4 unique J
Count
Nucleotide Difference

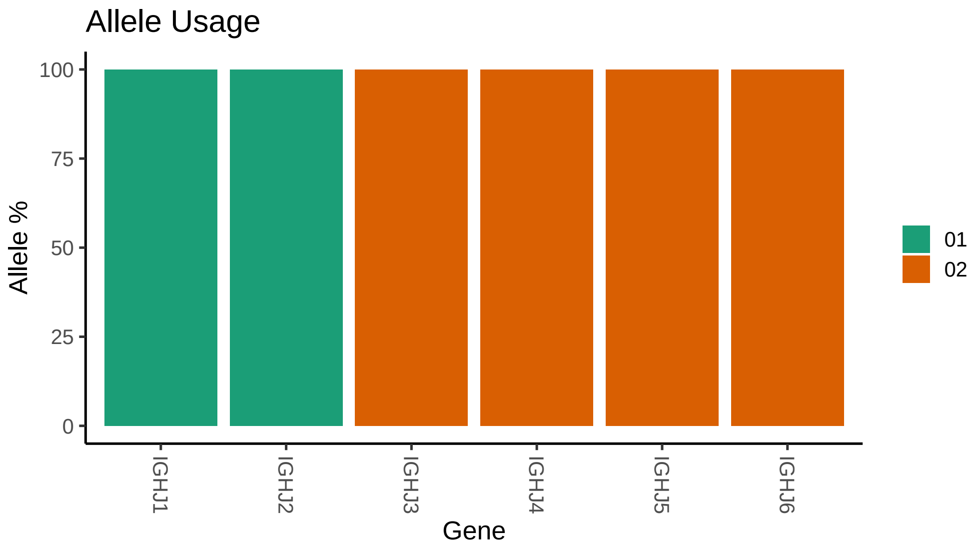

IGHV3−69−1*02
89 sequences assigned
2 (2.2%) exact matches, in which:
2 unique CDR3
2 unique J
IGHV3−72*01
1464 sequences assigned
57 (3.9%) exact matches, in which:
55 unique CDR3
4 unique J
IGHV3−73*01_02
1308 sequences assigned
402 (30.7%) exact matches, in which:
379 unique CDR3
5 unique J
IGHV3−74*01
7470 sequences assigned
1005 (13.5%) exact matches, in which:
889 unique CDR3
6 unique J
IGHV3−74*01_A315G
4673 sequences assigned
231 (4.9%) exact matches, in which:
230 unique CDR3
6 unique J
IGHV3−43D*04_05
279 sequences assigned
117 (41.9%) exact matches, in which:
113 unique CDR3
5 unique J
IGHV3−64D*06
8 sequences assigned
No exact matches.
IGHV3−64D*08
5 sequences assigned
No exact matches.
IGHV3−NL1*01
198 sequences assigned
No exact matches.
IGHV4−4*02_03
4425 sequences assigned
1690 (38.2%) exact matches, in which:
1623 unique CDR3
6 unique J
IGHV4−4*02_03_A245C_C300T
820 sequences assigned
194 (23.7%) exact matches, in which:
194 unique CDR3
6 unique J
IGHV4−28*03
31 sequences assigned
No exact matches.
Count
Nucleotide Difference


IGHV4−28*01_07
15 sequences assigned
5 (33.3%) exact matches, in which:
5 unique CDR3
3 unique J
IGHV4−28*02_05
48 sequences assigned
No exact matches.
IGHV4−30−2*01
4237 sequences assigned
756 (17.8%) exact matches, in which:
696 unique CDR3
6 unique J
IGHV4−30−4*01
4785 sequences assigned
1216 (25.4%) exact matches, in which:
1169 unique CDR3
6 unique J
IGHV4−30−2*01_C315G
703 sequences assigned
169 (24%) exact matches, in which:
169 unique CDR3
5 unique J
IGHV4−30−4*01_C315G
1618 sequences assigned
177 (10.9%) exact matches, in which:
177 unique CDR3
6 unique J
IGHV4−30−2*01_G245C_C300T_C
367 sequences assigned
96 (26.2%) exact matches, in which:
95 unique CDR3
3 unique J
IGHV4−31*03
5356 sequences assigned
1525 (28.5%) exact matches, in which:
1468 unique CDR3
6 unique J
IGHV4−34*01
14497 sequences assigned
6987 (48.2%) exact matches, in which:
6532 unique CDR3
6 unique J
IGHV4−34*01_T300C
5556 sequences assigned
529 (9.5%) exact matches, in which:
528 unique CDR3
6 unique J
IGHV4−38−2*01
2363 sequences assigned
847 (35.8%) exact matches, in which:
823 unique CDR3
6 unique J
IGHV4−38−2*01_A291G_C300T
421 sequences assigned
144 (34.2%) exact matches, in which:
144 unique CDR3
5 unique J
Count
Nucleotide Difference


IGHV4−38−2*01_C288T_A291G
296 sequences assigned
89 (30.1%) exact matches, in which:
89 unique CDR3
5 unique J
IGHV4−39*07
1354 sequences assigned
5 (0.4%) exact matches, in which:
5 unique CDR3
2 unique J
IGHV4−39*01_08
4152 sequences assigned
1213 (29.2%) exact matches, in which:
1153 unique CDR3
6 unique J
IGHV4−39*07_A103G_A106G_G11
1045 sequences assigned
113 (10.8%) exact matches, in which:
113 unique CDR3
5 unique J
IGHV4−59*01
10854 sequences assigned
4603 (42.4%) exact matches, in which:
4341 unique CDR3
6 unique J
IGHV4−59*12
2278 sequences assigned
227 (10%) exact matches, in which:
227 unique CDR3
6 unique J
IGHV4−59*12_C300T
2202 sequences assigned
916 (41.6%) exact matches, in which:
906 unique CDR3
6 unique J
IGHV4−61*01_i04
1732 sequences assigned
753 (43.5%) exact matches, in which:
733 unique CDR3
6 unique J
IGHV4−61*01_i04_G88A_A106G
583 sequences assigned
81 (13.9%) exact matches, in which:
81 unique CDR3
5 unique J
IGHV4−61*01_i04_G88A_A106G_(
661 sequences assigned
90 (13.6%) exact matches, in which:
90 unique CDR3
6 unique J
IGHV4−61*01_i04_T288C_G291A
436 sequences assigned
103 (23.6%) exact matches, in which:
101 unique CDR3
5 unique J
IGHV4−NL1*01
4770 sequences assigned
1641 (34.4%) exact matches, in which:
1562 unique CDR3
6 unique J
Count
Nucleotide Difference


IGHV4−NL1*01_C300T
637 sequences assigned
179 (28.1%) exact matches, in which:
178 unique CDR3
6 unique J
IGHV5−10−1*01_03
7 sequences assigned
2 (28.6%) exact matches, in which:
2 unique CDR3
2 unique J
IGHV5−51*01_03
9122 sequences assigned
3757 (41.2%) exact matches, in which:
3285 unique CDR3
6 unique J
IGHV5−51*01_03_A263T
4668 sequences assigned
1963 (42.1%) exact matches, in which:
1749 unique CDR3
6 unique J
IGHV6−1*01
1894 sequences assigned
467 (24.7%) exact matches, in which:
442 unique CDR3
5 unique J
IGHV7−4−1*02_T63C
6287 sequences assigned
2134 (33.9%) exact matches, in which:
1715 unique CDR3
6 unique J
IGHV7−81*01
1 sequences assigned
No exact matches.
Count
Nucleotide Difference

3 Allele usage in potential haplotype anchor genes



4 Haplotype plots

5 Configuration settings

```
## Repertoire file: /misc/work/jenkins/PRJNA491287/M64 105/M64 105/M64 105/M64 105/5b/b58e5b4b62109bc61
##
## Germline reference file: /misc/work/jenkins/PRJNA491287/M64 105/M64 105/M64 105/M64 105/5b/b58e5b4b6
##
## Novel allele file: /misc/work/jenkins/PRJNA491287/M64 105/M64 105/M64 105/M64 105/5b/b58e5b4b62109bc
##
## Species: Homosapiens
##
## Chain: IGHV
##
## Segment: V
##
## Unexpected N in reference sequence IGHV1 3 01_05: replacing with gap
##
##
## Unexpected N in novel sequence IGHV1 3 01_05_A291G_T300C: replacing with gap
##
##
## Unexpected N in reference sequence IGHV1 24 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV1 24 01_A315G_C317G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 11 04: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 11 04_T210C_G240A_C246T_G247T_T256A_A258G_C288T: replacing with
##
##
## Unexpected N in reference sequence IGHV3 13 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 13 01_A315G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 30 03: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 30 03_T288C: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 30 20_G75C: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 30 3 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 30 3 01_T288C: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 33 01: replacing with gap
```

```
##
##
## Unexpected N in novel sequence IGHV3 33 01_C288T: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 33 08: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 33 06_G75C_T300C_G303A: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 48 03: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 48 03_T303G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 11 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 48 04_C90T_T105C_A112G_G113C_A119G_A184G_A240G_T300C: replacing
##
##
## Unexpected N in reference sequence IGHV3 53 01_02: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 53 05_T267G_C300T: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 66 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 66 02_C223A_T267G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 74 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 74 01_A315G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 4 02_03: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 4 02_03_A245C_C300T: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 30 2 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 30 2 01_C315G: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 30 2 01_G245C_C300T_C315G: replacing with gap
```

```
##
##
## Unexpected N in reference sequence IGHV4 30 4 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 30 4 01_C315G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 34 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 34 01_T300C: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 38 2 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 38 2 01_C288T_A291G: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 38 2 01_A291G_C300T: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 39 07_A103G_A106G_G118A_A163G_G164A_T165A_T169A_T172C_G189A_T19
##
##
## Unexpected N in reference sequence IGHV4 59 12: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 59 12_C300T: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 61 01_i04: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 61 01_i04_T288C_G291A: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 59 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 61 01_i04_G88A_A106G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 31 03: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 61 01_i04_G88A_A106G_G129C_C134A_A147C_T165C: replacing with ga
##
##
## Unexpected N in reference sequence IGHV4 NL1 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 NL1 01_C300T: replacing with gap
```

```
##
##
## Unexpected N in reference sequence IGHV5 51 01_03: replacing with gap
##
##
## Unexpected N in novel sequence IGHV5 51 01_03_A263T: replacing with gap
```